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DROPT: Dynamic Rank Order Process Tracing

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Abstract

Ranking is ubiquitous and is widely used by researchers to measure preferences. The standard implementation of ranking offers limited insight into how those ranks are constructed. We present Dynamic Rank-Order Process Tracing (DROPT), a simple JavaScript-based method that records how individuals drag and drop items into rank-order positions, and provide a detailed tutorial to facilitate its use. Across three studies, we illustrate how DROPT records ranking behavior over time and can support inferences about attribute use in rank-order responses. In Studies 1A and 1B, participants ranked perceptual items by objective attributes (e.g., number of corners). Participants consistently showed a “3F” pattern, moving items scoring higher on the task-relevant attribute more *frequently*, *first*, and *farther* upward. In Study 2, we replicated the 3F pattern using lotteries that varied along the dimensions of probability of winning and payoff. We further show that the observed 3F pattern is more consistent with a ranking process resembling a selection sort than with several alternative sorting strategies. We also find that drag and drop patterns are associated with preferences: Participants who opted for lotteries with higher probabilities moved these lotteries more frequently, first and farther up, while the opposite was true for participants who opted for lotteries with higher payoffs. Incorporating information of the ranking process captured by DROPT improved the prediction of preferences in subsequent tasks beyond the standard case of observing final rankings alone. Together, we present a novel rank-specific implementation and analytic framework for collecting and analyzing drag-and-drop paradata in rank-order tasks.

Keywords: Process-tracing, tutorial, ranking data, preference construction, prediction

DROPT: Dynamic Rank Order Process Tracing

Ranking is ubiquitous in modern life. Sports teams are ranked in a top 25, even if they have not directly competed. There are competing top 50 and top 100 lists of the best restaurants in the world. In the film *High Fidelity*, John Cusack and friends endlessly argue over his top 5 romantic breakups and other rankings.

Ranking is deceptively simple. Each entry is represented by a simple position in an ordered list. Yet, this ranking often represents an aggregate evaluation across several different properties. Football rankings represent the number of wins, margins of victory, the strength of the opposing schedule, and offensive and defensive performances. The top 50 restaurants may juxtapose an elegant restaurant with a multi-course tasting menu with a romantic seaside fish specialist in a beachfront hut.

Ranking can be understood as a complex process of trading off different properties of diverse options. To rank two options, one must decide if the differences on one attribute exceed the weaknesses on another. For example, does the fresh-caught seafood compensate for the lackluster restaurant design? Understanding this tradeoff process can deliver insight into the nature of preferences.

In this paper, we develop a novel method for gathering rank process data that can record observable traces of how rankings unfold over time. We also provide a step-by-step tutorial on how to implement our process-tracing method in Qualtrics, a widely used survey software. We call this method “Dynamic Rank-Order Process Tracing” (or DROPT). While final rankings hold important information about preferences, we also seek to study the *process* of ranking, by recording the time and position of items being ranked, and their changes over time. Here, we use DROPT to provide an initial demonstration of how movement data can inform inferences about

rank-order judgments and to examine how these process data can be used for prediction.

Importantly, we do not claim that movement data directly measure attention or cognition; rather, we treat them as process indicators that may provide insight into how rankings are constructed under the assumption that the item being dragged is the option currently under consideration (Costa-Gomes et al., 2001).

DROPT is a technique designed to record interaction data generated during drag-and-drop rank-order tasks. While prior work has examined the effects of ranking information on decision making (Chun & Larrick, 2022; Weingarten et al., 2023), surprisingly little research has focused on the process by which people generate rankings in standard online interfaces. DROPT enables researchers to analyze intermediate steps within ranking processes. This includes, for instance, the order in which study participants reorganize items to come up with their final rankings, how much time they spend between ranking steps, how often each individual item is touched, or the distance each item travels before it arrives at its final position. The method records paradata—“digital traces left by respondents” (Vehovar et al., 2024, p. 225)—such as mouse clicks or touch events (McClain et al., 2019).

DROPT has several advantages for researchers studying decision-making processes. First, DROPT is easy to implement, as we will show in the upcoming step-by-step guide, and can be deployed at scale with minimal additional costs. Specifically, DROPT does not require specialized software beyond a survey platform (in this tutorial we use Qualtrics) and R; the main added cost is the initial setup of the Qualtrics question, JavaScript logger, and analysis script. Using DROPT in its standard form requires only basic familiarity with Qualtrics, JavaScript, and R, while more advanced users can adapt the code to their own purposes. Second, dynamic rank data are recorded in the background of online surveys, allowing researchers to collect high-

resolution process data without imposing additional cost, as in concurrent verbal reports. Third, DROPT is highly flexible. Because it is built in JavaScript, it can be easily customized to different study designs, addressing various research questions about how rankings are constructed. Finally, our method provides a novel, rich source of data that can be used to model and validate inferences about rank-ordered multi-attribute judgments. To the best of our knowledge, DROPT is the first rank-specific implementation and analytic framework for collecting and analyzing drag-and-drop paradata in online rank-order tasks.

Literature Review

Ranking Processes

We define *ranking* broadly as ordering a set of options along a criterion, whether by an objective property (e.g., shapes from smallest to largest in size) or a subjective property (e.g., one’s preferences over lottery options from most to least). Rank data is often gathered using “drag-and drop” online interfaces where people select an item, drag it to a new position, and drop it into its new position in the list by releasing their finger or cursor. For example, consumers may reorder songs using drag-and-drop in a Spotify playlist. In research settings, behavioral scholars have used this format to study individual knowledge and preferences (see Johnson et al., 2024; Ludwig et al., 2025). In fact, Blasius (2012) identified the drag-and-drop method as a particularly well-suited format to collect rank data in web surveys. We employ this broadly familiar interface, asking individuals to rank-order a pre-populated list in an online survey, while recording information about their drag-and-drop moves in the background.

Could such data inform inferences about ranking processes, such as attribute usages and trade-offs? The decision-making literature suggests that choices and judgments are often constructed in the moment, and uncovering the deliberation process may reveal the unfolding of

psychological processes over time (Fischhoff & Broomell, 2020; Schulte-Mecklenbeck et al., 2017, 2019). For instance, eye-tracking studies use eye fixations to infer patterns of information search (Ludwig et al., 2024; Martinovici et al., 2023; Orquin & Mueller Loose, 2013; Russo & Rosen, 1975) and mouse-tracking studies analyze cursor trajectories and movements paths to capture attentional processes (see Stillman et al., 2018, for a review). These approaches share the assumption that observable behaviors, such as eye fixation, box opening, or cursor position, reflect the focus of consideration (Schulte-Mecklenbeck et al., 2017, 2019). In particular, the adjacency assumption in mouse-tracking research suggests that the temporal order of observable actions can provide evidence about the sequence of information processing (Willemsen and Johnson, 2011; Costa-Gomes et al., 2001).

In ranking tasks, we make a parallel assumption: The item being dragged and dropped at any given moment reflects the option people are actively considering. Thus, although we do not claim that DROPT is a direct measure of attention, whether an item is moved, when it is moved, and how far and in which direction it is moved may provide indicators of which options participants are considering and how those considerations unfold over time. See also recent work by Brucks and Levav (2026) linking attention and dragging movements in online interfaces.

While many metrics can be derived from drag-and-drop data, we focus on three item-level properties. First, *drag count* captures how *frequently* an item is moved. Second, *drag distance* captures how *far* the item is moved. Finally, *drag order* captures when each movement occurs in the sequence of movements - for example, whether an item is moved first, last, somewhere in between, or not at all. We define each metric in greater detail later.

Sorting Strategies

One reason to explore these three metrics is that they help us understand the underlying process. When asked to rank shapes from the most to least number of corners, how might respondents approach this task? Consider respondents presented with a randomized order of the items in the first row of Figure 1. How might respondents drag and drop the options to reflect the number of points (vertices) of each object?

The computer science literature contains many suggested sorting algorithms (Knuth, 1998) and which we use as a starting point. Respondents might begin by asking themselves which option has the greatest number of corners, identify the respective item, and move it to the top of the list. They may continue, asking which of the remaining options has the most corners among the remaining options and place that item second. This process resembles selection sort, a standard sorting algorithm in the computer science literature in which one iteratively identifies the highest (or lowest)-ranked item from an unsorted list and places it into its correct position, then repeats the process on the unsorted list.

While selection sort is just one possible ordering strategy, it predicts a unique pattern of the three characteristics we use to describe the drag and drop data. Because individuals repeatedly identify and move the option that best satisfies the comparison, items with higher values of the relevant attribute are moved earlier. With random starting positions, these items, on average, will be moved (i) more frequently, (ii) first (i.e., earlier), and (iii) farther upward. If executed without error, each object is moved, at most once. The empirical data was consistent with this, and we therefore operationalize “frequently” as whether an item was moved at all. Together, selection sort yields what we call a “3F” pattern: Items with higher values of the attribute are moved more frequently, first, and farther toward the top as reflected by our metrics of drag count, drag order, and drag distance.

A mirror image of this strategy would proceed from the lowest to the highest value on the underlying attribute, producing the opposite prediction: The option with the highest value of the substituted attribute would, on average, be moved *least* frequently, *last*, and *least* distances (a "3L" pattern).

Compare this to another commonly discussed sorting procedure, bubble sort (Astrachan, 2003; Knuth, 1998). Here one starts at the top of the list, compares adjacent pairs, swaps items if needed, and repeats this process recursively until the list is fully ordered. Under a bubble sort strategy, items under consideration would typically be dragged up or down by only one position at a time, but would be dragged repeatedly, in contrast to a selection sort. We will offer a more formal analysis shortly.

Each sorting strategy comes with different trade-offs. The bubble sort, for instance, involves a minimal memory load since two options are considered at a time. But it usually requires many comparisons. While the computer science literature focuses on which algorithm is optimal under different definitions of search costs and input patterns, that is not our focus. Instead, we simply want to use the observed drag and drop data to characterize the observed process.

In our setting, participants are instructed to rank items from top to bottom, reflecting a conventional format for ranking tasks in everyday contexts (e.g., "top choice" lists or preference orderings). While we acknowledge that other ranking strategies are likely to occur in other tasks, and we encourage readers to adopt the tool to explore alternatives, we find that the selection sort strategy provides an adequate account of drag and drop behavior in the tasks we examine.

Our evidence shows widespread evidence that drag-and-drop data is consistent with a selection sort because we most often see, the highest valued item dragged first. Across a variety

of starting orders, the highest-valued item was most often moved first. Selection sort also predicts that higher-valued items will be dragged farther and moved more frequently. In sum, we test whether participants' movement patterns are consistent with a strategy in which they iteratively and implicitly ask, "which option has the highest value of attribute X (among the items that have not been moved)?" and placing that option in the right position next, a strategy that produces the 3F pattern. Our simulations show that alternative ranking strategies do not produce the same pattern.

Building on the assumption that the item being dragged reflects the option currently under consideration, examining this ranking process can provide new insights into how various judgments are constructed (Payne et al., 1993). We can extend this analysis to items that differ on different attributes, and can potentially learn which attributes people use, how they prioritize or weigh different attributes, and even how strategies evolve over time—information that the final ranking alone cannot capture (Bettman et al., 1998; Chun & Larrick, 2022).

The remainder of the paper illustrates the method through three empirical examples, including a step-by-step guide for implementing the tool and analyzing the resulting data.

Overview of Studies

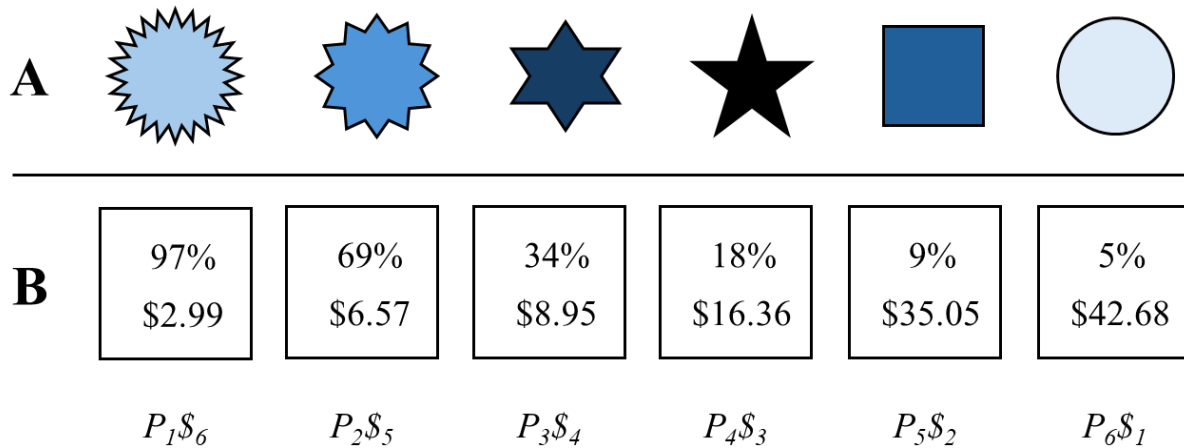
Across three studies, we illustrate the type of data generated by DROPT and examine how it can inform the study of underlying preferences. First, in Study 1A, we provide an initial demonstration of DROPT using rank-order tasks where there are objectively correct answers. We benchmark our results to a set of alternative sorting strategies. Then, in Study 1B, we provide a conceptual replication of Study 1A and, at the same time, walk the reader through the implementation of DROPT and the data analysis process. In this walkthrough, we explain how to implement DROPT in Qualtrics and analyze the resulting data step-by-step, as well as how to use

a DROPT-specific R function. Finally, in Study 2, we move to a domain that allows for subjective preferences and demonstrate the added out-of-sample prediction accuracy that DROPT can provide.

Figure 1 previews the stimuli used across studies. Panel A shows the geometric shapes used in Studies 1A and 1B, which varied in color shading and number of corners. Panel B shows the lotteries used in Study 2, which varied in probability of winning and payoff. We label these lotteries with P for probability and \$ for payoff. A subscript indicates the corresponding rank, with 1 being the highest probability, or highest payoff, and 6 the lowest. So, $P_1\$_6$ denotes the gamble with the highest probability of winning and the lowest payoff.

Figure 1

Stimuli in Studies 1A and 1B (Panel A) and Study 2 (Panel B)



Note. In Panel A, the geometrical shapes are ordered by number of corners, from most (left) to least (right); In Panel B, the lotteries are ordered by probability of winning from highest (left) to lowest (right). The italicized labels underneath the lotteries indicate the ranking by probability (P) and payoff (\$), with 1 being the highest probability, or highest payoff, and 6 the lowest.

Because the three studies shared recruitment procedures and several task elements, we describe these common methods in the following sub-sections “Participants” and “Design and Procedure” before turning to study-specific details and analyses.

Participants

We recruited participants with a current residence in the United States using the online panel provider Prolific. Participation was restricted to individuals accessing the study on a desktop or laptop computer. Participants were paid \$1.00 (Studies 1A and 1B) and \$2.00 (Study 2) for completing our studies that took 8 (Studies 1A and 1B) and 14 (Study 2) minutes, respectively. We aimed to recruit 100 participants for Studies 1A and 1B, and 200 participants for Study 2. See Table 1 for an overview of studies. Participants were not permitted to take part in more than one study.

Table 1

Overview of Studies

Study	Description	Participants	Pre-Registration
1A	Participants rank-ordered shapes by number of corners and color.	45 female, 49 male, $M_{\text{age}} = 40.12$ ($SD = 13.23$)	https://aspredicted.org/7j5s-sc9h.pdf
1B	Conceptual replication of Study 1A with reversed instructions.	45 female, 46 male, 1 non-binary, $M_{\text{age}} = 36.34$ ($SD = 10.83$)	https://aspredicted.org/zqhx-96pp.pdf
2	Participants ranked lotteries by probability to win, payoff, and preference. They repeated the preference ranking for another set of six lotteries (incentivized) and indicated their preferred lottery in binary choice tasks.	93 female, 98 male, 3 non-binary, $M_{\text{age}} = 27.28$ ($SD = 11.48$)	https://aspredicted.org/tt62-mkfq.pdf

Note. Participant counts reflect sample sizes after excluding those who failed an attention check and whose responses were flagged as bots by Qualtrics. Although these exclusions were not preregistered for Studies 1A–1B, they were applied for consistency with Study 2 and minimally affected the results. Additional exclusions were applied on a task-level (i.e., ranking by color, etc.) prior to the analysis of each task.

Design and Procedure

The three studies shared various elements, which are elaborated in more detail below. A schematic representation of study procedures can be found in Figure SM1 in the Supplementary Materials (SMs).

Warm-up Task

Participants in all studies started with a practice task to familiarize themselves with the rank-order question format. In the task, participants were asked to rank-order five circles enclosed by squares from largest (at the top) to smallest (at the bottom) circle area. Starting positions in this and all subsequent tasks were randomized.

Rank-Order Tasks

In Study 1A, each participant rank-ordered the set of six geometric shapes shown in Figure 1A in two ways. Participants rank-ordered these shapes by color, from darkest (top) to lightest (bottom), and by number of corners, from highest (top) to lowest (bottom). We counterbalanced the order of tasks between participants. The two attributes—color and number of corners—were uncorrelated ($r = -0.02$).

In Study 1B, participants ranked the same shapes with reversed instructions: Placing the lightest shape or the fewest-corner shape at the top. We again counterbalanced the task order.

In Study 2, participants rank-ordered six lotteries (see Figure 1B) across three blocks with three different instructions, ranking by probability of winning, payoff, and preference, with randomized order of blocks. In each task, participants were instructed to rank lotteries from

highest to lowest on the relevant dimension, that is, probability of winning, payoff amount, or their own preference. We refer to the ranking by probability and ranking by payoff blocks as *attribute ranking tasks*. These blocks represented a conceptual replication of Study 1, as they instructed participants to rank items by an objective attribute. We refer to the third block, measuring subjective preferences, as the *preference ranking task*. As we will explain in more detail below, this block served to investigate the relationship between DROPT process data and preferences constructed in a multi-attribute setting. A fourth block, the *validation ranking task*, introduced six new lotteries that participants ranked according to their preferences (see Table SM4 for stimulus details). To incentivize this task, we randomly selected two lotteries, played the higher-ranked one, and paid out winnings. Probabilities and payoffs were jittered while preserving their rank orderings, and all lotteries had an approximately equal expected value (\$2.83). Participants also indicated whether they believed they could win a bonus.¹

Binary Choice Tasks

Finally, in Study 2, participants made five binary choices between two lotteries in a fifth block, the *validation choice task* (see Table SM7 for stimulus details). We randomized the order of these lottery sets. We use the validation ranking and binary choice tasks to test if the process data from the preference task can capture elements of preferences beyond final choices alone.

Recall and Attention Check

In each study, we asked participants to recall the ranking tasks they had just completed (see *Study 1 Supplementary Methods* for details). Additionally, we asked as an attention check whether participants worked 28 hours in a typical day and excluded those who responded with yes or unsure.

¹ “Jittered” here means that displayed probabilities and payoffs were randomly varied within small prespecified ranges around each lottery’s base values. See Table SM4 for the jitter amounts by lottery and additional details.

Demographics and Additional Questions

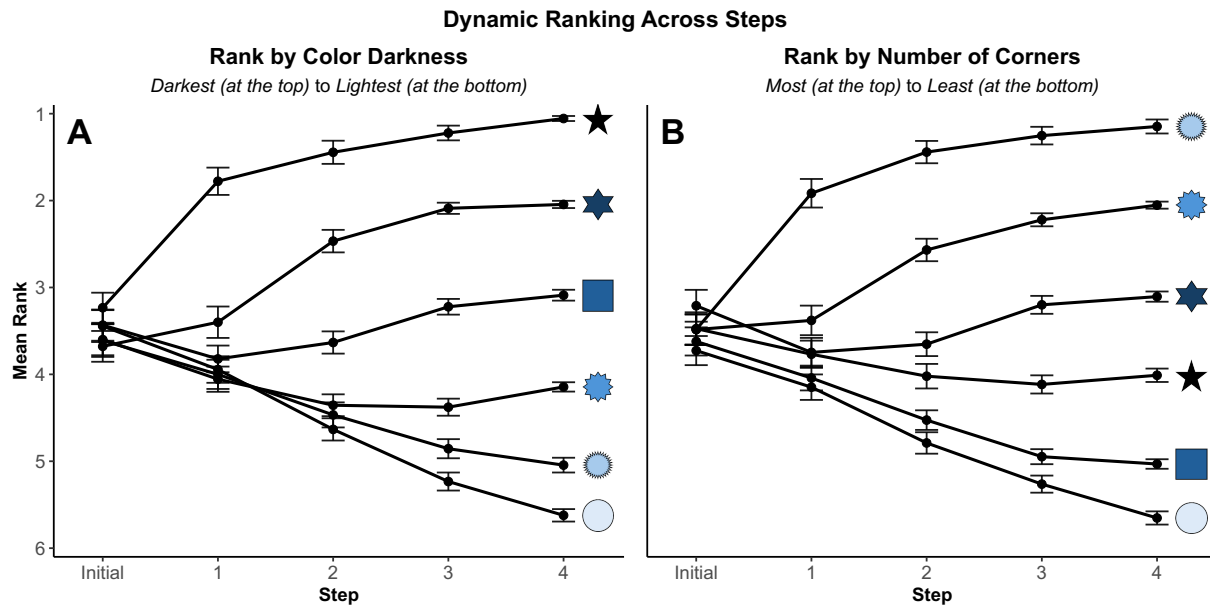
Participants indicated the device they used to complete the ranking task (e.g., mouse, touchscreen, etc.). In Study 2, we also asked participants for their risk preference on an 11-point scale anchored at “not at all” to “very willing” to take risks (Dohmen et al., 2011). Finally, we asked participants for demographic information and additional debrief questions, including if they found the study tedious or confusing (see *Study 1 Supplementary Methods* for details).

Study 1: Ranking Geometric Shapes

Study 1A

Respondents were instructed to rank shapes either by color shading (from darkest to lightest) or by the number of corners (from most to fewest). As preregistered, before analyzing the attribute ranking tasks, we applied two additional (task-level) exclusion criteria. We removed individuals with data recording errors as well as those who exhibited substantial deviations from the correct ranking. Specifically, we calculated task accuracy for each participant using Kendall’s rank correlation coefficient (τ), which measures the correspondence between a participant’s ranking and the objectively correct ranking ($\tau = 1$ indicates a perfectly correct ranking; $\tau = -1$ indicates a perfectly reversed ranking). We exclude participants whose τ is less or equal to 0, indicating rankings that were at chance ($\tau = 0$) or reversed ($\tau < 0$) relative to the correct ordering. We obtained a final sample of 87 participants in the color task and 89 participants in the corner task. Across tasks, respondents made an average of almost four drags and drops per task, ($M_{\text{color}} = 3.91$, $SD = 1.18$; $M_{\text{corner}} = 3.92$, $SD = 1.33$), and completed the tasks accurately ($M_{\tau[\text{color}]} = 0.99$, $SD = 0.06$; $M_{\tau[\text{corner}]} = 0.98$, $SD = 0.09$), suggesting high levels of task engagement and comprehension.

DROPT enables us to capture descriptive patterns of rank changes over time. These are shown in Figure 2, broken down into steps of adjusting the ranking. Each step reflects the resulting rank-order state after an active drag-and-drop action; hence, these trajectories include both active movements of dragged items and passive shifts of items that mechanically change positions when other items were inserted above or below them. Panel A shows the average ranks for each step taken by participants in the ranking-by-color task. The darkest shape climbed up the ranks most quickly. There was a steep increase in mean rank between the initial position and step 1. At the next step, the second-darkest shape, now the highest-scoring among the remaining items, rose most quickly between step 1 and step 2. Low- to mid-scoring shapes shifted more gradually in rank. In the Qualtrics ranking interface, items are automatically moved downward as others are placed above them. These more gradual shifts were likely accounted for by these passive mechanical dynamics. By ranking step 4, most respondents had converged on the correct rank order, as indicated by the small standard errors. We observed the same patterns for ranking by corners (Figure 2B), where the shape with the most corners was moved up to the first position first, etc. These patterns suggest that participants, on average, tended to initially move the highest-scoring items upward and then proceed iteratively to the next-highest options. Under such an approach, each item required only one adjustment. Moreover, lower-scoring items gradually and mechanically shifted downward as higher-scoring items were positioned above them.

Figure 2*Steps of Ranking Shapes by Color (A) and Corners (B) in Study 1A*

Note. Mean rank of each shape across drag-and-drop steps in color darkness (Panel A) and number of corners (Panel B) tasks. Error bars represent ± 1 standard error across respondents at each step. The first four drag-and-drop steps are displayed. Mean rank changes reflect both active movements and passive rank shifts of items.

The plots in Figure 2 describe the evolution of mean ranks over ranking steps. We next conducted a more granular analysis using the process data generated by DROPT, considering how often each item was actively moved (drag count), when it was actively moved (drag order), and how far it was actively moved (drag distance; positive values of drag distance indicate an upward move in the ranking). Because the vast majority of items (around 95%) were moved either once or not at all, we used an indicator variable for drag count (0 for unmoved, 1 for moved), and drag order and drag distance were calculated based only on the first drag-and-drop action, which captures the only movement in the vast majority of cases. For drag distance, positive values indicate upward movement. Unmoved items were assigned a drag count of 0, a

drag distance of 0, and a drag order equal to the maximum observed drag order in the task plus 1 (e.g., if four items were moved, all unmoved items were assigned a drag order of 5). We note that the three measures are conceptually separable. Drag count captures whether an item was moved at all, but a moved item can be moved early or late, and it can be moved a short or long distance upward or downward. Similarly, drag order captures when an item was moved, but items moved at the same point in the process may differ substantially in how far they were moved and in which direction.

Using the three measures, we tested how task-relevant and task-irrelevant attributes influenced ranking behaviors. We combined data across both tasks and included four nested predictors.² These included (1) two nested relevant attributes, where the attribute matched the task (color darkness in the color task, number of vertices in the corner task) and (2) two nested irrelevant attributes, where the attribute did not match the task (number of corners in the color task, color darkness in the corner task). For each dependent measure (drag count, drag order, and drag distance), we fit separate regression models that included the four predictors. All models controlled for the initial item positions and addressed dependencies in the data arising from repeated measures (see *Study 1 Supplementary Results, Regression Analyses for 3F Patterns* for more details).

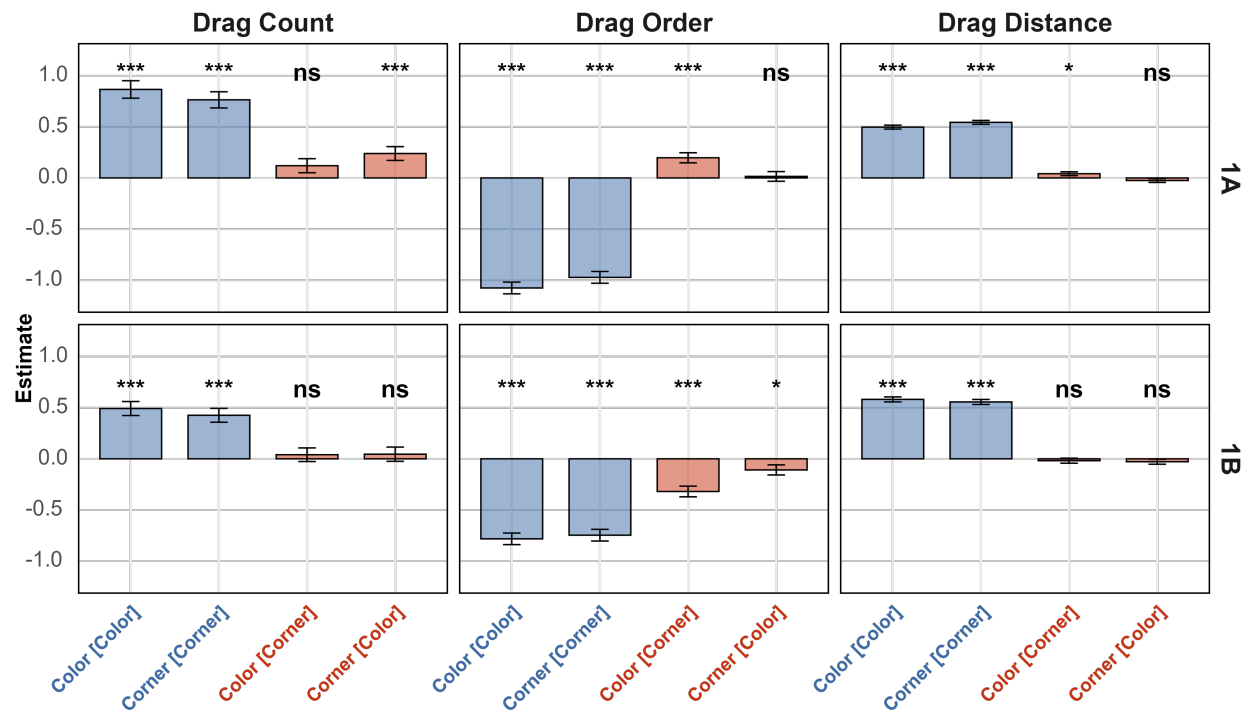
The results aligned with the 3F pattern in both tasks. As Figure 3 shows, relevant predictors (blue bars) significantly predicted all three measures. For drag count, coefficients were significantly positive, indicating that items with higher attribute values were more likely to be

² We combined data across tasks to improve power and conciseness; separate task-level analyses produced substantively consistent results. Our preregistered specification produced singular-fit warnings for some outcomes, suggesting overparameterization. We therefore compared the preregistered model with simpler specifications that dropped random effect terms and reported the best-fitting estimable specification in the main text. Coefficient estimates were similar across specifications. See the *Regression Analyses for Testing the 3F Pattern* section of the SMs for details.

moved more often. For drag order, coefficients were significantly negative, indicating that higher-value items were moved earlier in the ranking process. For drag distance, coefficients were significantly positive, indicating that higher-value items were moved farther upward toward the top of the list.

Figure 3

Regression Results of Drag-and-Drop Behavior for Studies 1A-1B



Note. Model-estimated coefficients (± 1 SE) across the three drag-and-drop regression models. All predictors were mean-centered values and reverse-coded so that larger values indicate higher attribute values³. Each predictor refers to an attribute nested within a task, with the task indicated in brackets (e.g., “Color [Color]” indicates color darkness nested in the Color task). Relevant predictors are shown in blue and irrelevant predictors in red.

³A note on rank coding: In descriptive figures showing rank positions, ranks retain their conventional coding, with 1 indicating the highest or top position and 6 the lowest or bottom position. For regression analyses and coefficient displays, we reverse-coded the attribute-rank predictors so that larger values indicate higher attribute values (6 = highest; 1 = lowest). Under this coding, the 3F pattern appears as a positive coefficient for drag count, a negative coefficient for drag order, and a positive coefficient for drag distance.

*Asterisks above each bar denote significance: * $p < .05$, ** $p < .01$, *** $p < .001$; ns = not significant.

Do drag count, order, and distance capture the same underlying construct? They should be correlated if a selection sort is used: items that are moved are also likely to be moved earlier and farther upward. For drag distance, variation in initial positions can weaken its association with the other two measures, even if participants are using a systematic sorting strategy: a high-value item starting near the top may move only slightly, whereas the same item starting near the bottom may move much farther. In the analyses below, we control for initial position fixed effects to reduce this mechanical influence.

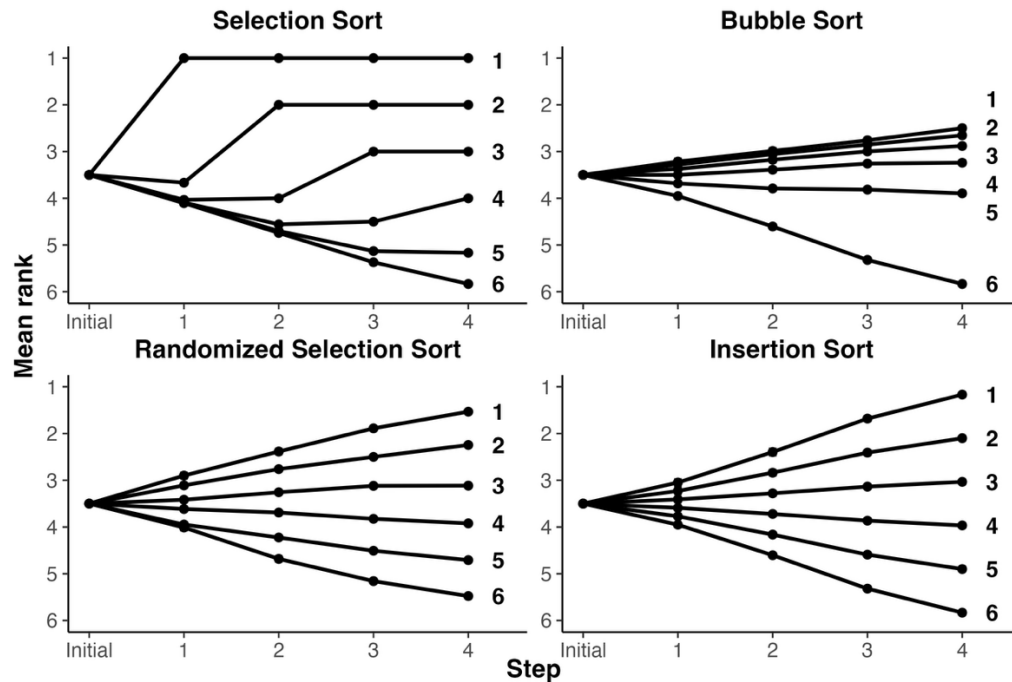
For each respondent, we calculated correlations between pairs of measures across items. To control for the starting positions, we calculated the residual for each measure with respect to initial position and added participant-level random intercepts prior to computing within-respondent correlations. This approach yielded three correlation coefficients per participant. We found moderate to high correlations in both the color ($M_{r[\text{count-order}]} = -0.73$, $SD = 0.20$; $M_{r[\text{count-distance}]} = 0.47$, $SD = 0.38$; $M_{r[\text{order-distance}]} = -0.74$, $SD = 0.29$) and the corner tasks ($M_{r[\text{count-order}]} = -0.73$, $SD = 0.21$; $M_{r[\text{count-distance}]} = 0.42$, $SD = 0.45$; $M_{r[\text{order-distance}]} = -0.61$, $SD = 0.43$), suggesting that the three measures captured overlapping aspects of the ranking process.

Simulation of Alternative Sorting Strategies

Is the 3F pattern unique to the proposed selection sort process, or can other sorting strategies that produce the correct final rankings generate similar patterns? To address this question, we conducted simulations of the proposed selection sort, along with three alternative strategies that participants might use: randomized selection sort, bubble sort, and insertion sort. Randomized selection sort preserves the idea that participants select and move an item to its correct position at a time, but assumes that they randomly select which item to move rather than

asking which remaining item has the highest value on the task-relevant attribute, providing a random baseline. In bubble sort, participants compare adjacent items and make swaps until the list is ordered. Finally, rather than searching the full list for the highest-value remaining item, participants may simply scan down the list and drag each item into its appropriate position among the sorted list. This is captured by insertion sort. Table SM3 illustrates the ranking process implied by each strategy.

We simulated each sorting strategy across all possible starting orders and compared the resulting movement patterns to the empirical pattern observed in Study 1A. As shown in Figure 4, the proposed selection sort most closely matched the empirical pattern in Figure 2: the highest-ranked item is moved to the top first, followed by the next-highest-ranked remaining items in order, which produces the same graded prioritization pattern observed in Study 1A. Alternative sorting strategies produce more distinct patterns. The simulation also suggests that the degree of overlap among drag measures depends on the ranking process being used. Additional comparisons reported in the *Simulation Results of Alternative Sorting Strategies* section of the SMs further suggest that participants' ranking behavior most closely resembled a selection sort strategy. The R code used to conduct these simulations is available on the Study 1 OSF page.

Figure 4*Steps of Ranking Shapes by Sorting Strategies*

Note. Each panel plots the simulated mean rank position of each item across all 720 possible initial orders. Numeric labels to the right indicate the rank of each item, with 1 denoting the highest-value item and 6 denoting the lowest-value item. Error bars are omitted because simulated standard errors were very small and visually indistinguishable in the figure. The first four drag-and-drop steps are displayed. The selection-sort process most closely mirrors the empirical 3F pattern observed in Study 1A.

Study 1B: A Guided Walkthrough

Do the patterns hold if participants are instructed to rank in reverse—that is, from lightest to darkest color or from fewest to most corners? When ranking from most to fewest corners, the shape with the most corners rises to the top early. But if asked to rank from fewest to most, would the smooth circle—the shape with the fewest corners—now be moved more often, earlier, and farther upward? We reversed the ranking instruction to test whether participants continue to

focus on the dominant end (e.g., more corners, darker color) even when asked to rank in reverse. This approach is related to the idea of markedness (Battistella, 1996): In many pairs (like “many-few” or “dark-light”), one side is the natural starting point, while the other takes more effort to think about (Clark, 1969; Lehrer, 1985). If markedness shapes ranking behavior, then reversing the instruction may lead to a mismatch rather than a clean flip: Darker items or items with more corners should still capture earlier touches and require more corrective moves to reach their final positions.

We also use this study as a guided working example of implementing DROPT step-by-step. We focus on the data collection software offered by Qualtrics (Qualtrics, 2025; <https://www.qualtrics.com/>) with subsequent data processing and analysis in the statistical programming language R (R Core Team, 2025). We note, however, that the underlying DROPT logic is not tied to these tools. The JavaScript-based approach can be adapted to other platforms that support custom scripting, and our materials may serve as a starting point for such implementations. Alongside our data cleaning script, we provide a general Qualtrics template (“DROPT_Survey_Template.qsf”), that users can simply upload into their accounts, as well as a data processing and analysis script in R (“DROPT_R_Script.R”) on the OSF (https://osf.io/vaycw/overview?view_only=f7fe74ff872546cf8fc6841c94ed93c4).

Pre-Data Collection: Implementation in Qualtrics

DROPT employs JavaScript to capture paradata associated with the ranking process, which is then stored in Qualtrics embedded data fields. In this section, we outline the two steps necessary to capture information about the ranking process.

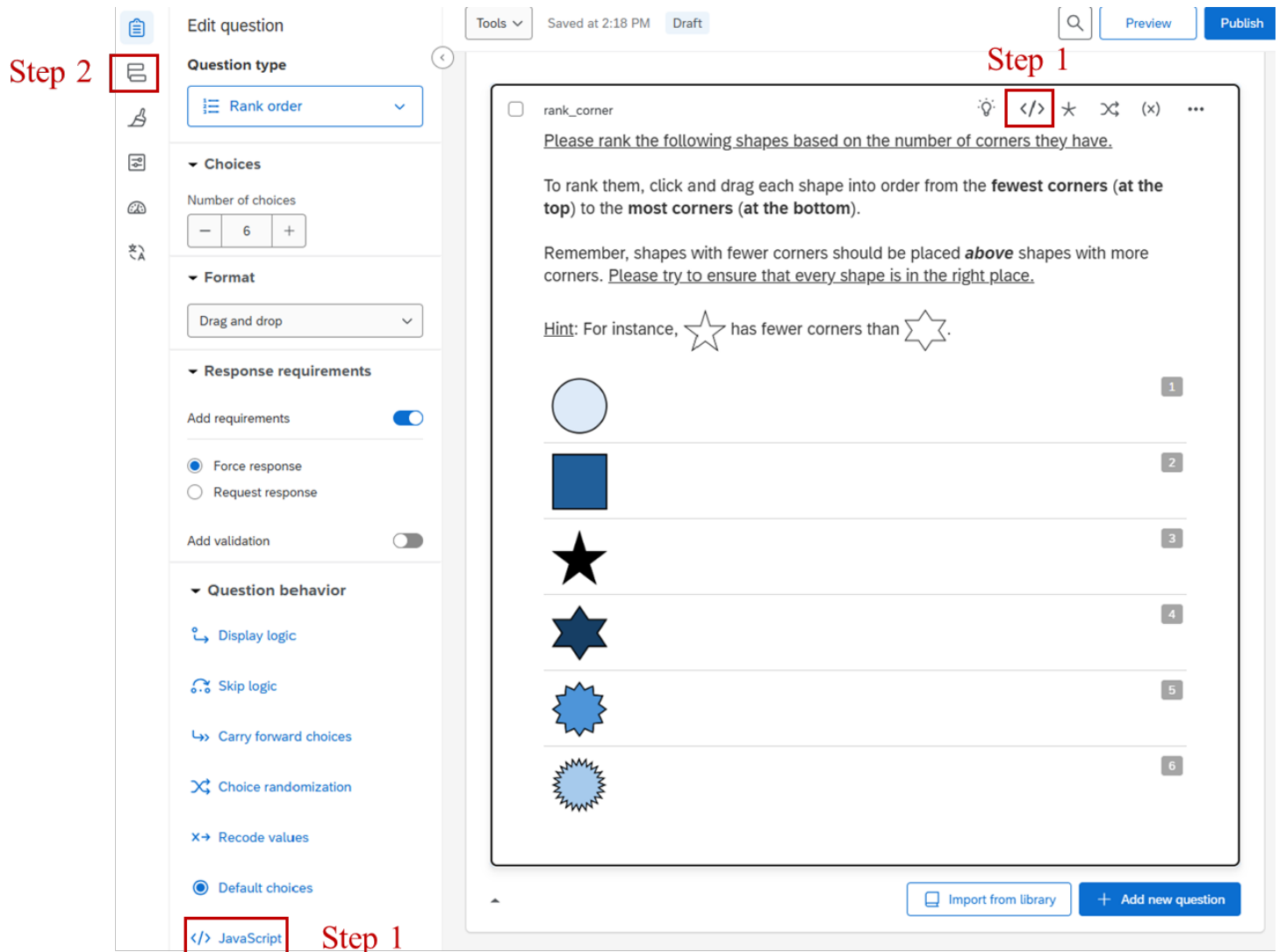
The first step is to embed the JavaScript in a rank-order question. This can be done by clicking on one of the JavaScript icons available for the rank-order question type, as highlighted

in Figure 5. The JavaScript is provided on the OSF (“DROPT_JavaScript.txt”) for readers to copy and adapt for their purposes. DROPT can be implemented for multiple rank-order questions within the same survey. To record paradata for each question, researchers can add the same JavaScript to every rank-order question for which DROPT measures should be collected. At each ranking step, the code records several types of information, including the item moved, timestamps when participants begin and end a step, the rank order after the adjustment, as well as task-level information that identifies the respective rank-order question (for details see the annotated JavaScript on the OSF). Because the final rank order alone does not uniquely identify which item was moved at a given step, the JavaScript also records which item was moved during a ranking task, together with the rank order after the move. To illustrate this point, consider a participant who rank-ordered three items in the following order (see also Figure 5): “circle”, “square”, and “black star”. If after one step the new order is “square”, “circle”, “black star”, we cannot tell from that information alone whether “circle” was moved *down*, or “square” was moved *up*. By capturing which item was moved, we can restore the ranking process.

The second step involves creating embedded data fields in the survey flow (see Figure 5 for navigation) in which the JavaScript can store the information. To record DROPT information, we create four embedded data fields: “DROPT_RankProcess”, “DROPT_TaskMetadata”, “DROPT_TaskCount”, and “DROPT_Submitted”. Each of these embedded data fields is automatically populated by the JavaScript when a participant completes the ranking task and becomes a variable in the downloaded data frame. We describe each variable in more detail in the next section. While experimenters can decide the specific naming of each embedded data field, the names must match the JavaScript to be recorded properly.

Figure 5

Screenshot of the Qualtrics Interface of the Rank-Order Question used in Study 1B



Note. Two steps required for implementing DROPT in Qualtrics, illustrated with an example from Study 1B.

Post-Data Collection: Data Processing and Analysis in R

Data can be downloaded from Qualtrics as numerical values and processed through a series of steps to derive the DROPT variables. For brevity, we do not reproduce the full code here but instead describe the main steps conceptually. However, readers can follow each step in

our tutorial script (OSF, “DROPT_5_Steps_Script.R”), which contains more detailed documentation. Readers who prefer a more compact workflow can instead use the fully integrated function script (OSF, “DROPT_R_Function.R”), which implements the same processing logic internally and returns the final item-level dataset in a one function call. Below, we describe the data structure, the conceptual steps to process and analyze DROPT data, the one-call R function, and discuss some researcher degrees of freedom when working with DROPT.

Figure 6

Exemplary Snippet of DROPT Variables

participant_ID	DROPT_RankProcess	DROPT_TaskMetadata	DROPT_TaskCount	DROPT_Submitted
ID93	{DROPT1 0 NA 4,3,5,1,2,6} {DROPT1 595,944 3 4,5,1,3,2,6} {D...	{DROPT1 QID5 4,3,5,1,2,6} {DROPT2 QID6 4,2,1,3}	2	1
UY33	{DROPT1 0 NA 6,1,5,4,3,2} {DROPT1 492,816 6 1,5,4,6,3,2} {D...	{DROPT1 QID5 6,1,5,4,3,2} {DROPT2 QID6 2,4,1,3}	2	1
LP19	{DROPT1 0 NA 1,5,2,4,3,6} {DROPT1 525,835 1 5,2,4,3,1,6} {D...	{DROPT1 QID5 1,5,2,4,3,6} {DROPT2 QID6 2,4,3,1}	2	1
WD31	{DROPT1 0 NA 4,3,2,1,5,6} {DROPT1 442,777 2 4,3,1,5,6,2} {D...	{DROPT1 QID5 4,3,2,1,5,6} {DROPT2 QID6 4,2,3,1}	2	1
QX36	{DROPT1 0 NA 5,1,2,4,6,3} {DROPT1 443,720 5 1,5,2,4,6,3} {D...	{DROPT1 QID5 5,1,2,4,6,3} {DROPT2 QID6 4,2,1,3}	2	1
FZ93	{DROPT1 0 NA 6,4,3,2,1,5} {DROPT1 462,792 6 4,3,2,1,6,5} {D...	{DROPT1 QID5 6,4,3,2,1,5} {DROPT2 QID6 3,1,2,4}	2	1

Note. Hypothetical data created post-data collection. The corresponding data structure in the presented studies differs slightly because the DROPT JavaScript and R Code were updated during the revision process.

Data Structure. When reading the data into R, the embedded data fields appear as separate columns, see Figure 6. The main processing information is stored in the column “DROPT_RankProcess.” If a survey contains more than one rank-order question, all rank-order tasks are written into this same field and are distinguished by chronological task IDs, such as “DROPT1”, “DROPT2”, and so forth. The column “DROPT_TaskMetadata” maps these task IDs back to the corresponding Qualtrics question IDs and item IDs.

Within “DROPT_RankProcess”, ranking steps are separated by curly brackets. Each entry has the following structure: {task_id|timing|item_moved|current_order}. The first entry for each task records the initial order at time “0” and uses “NA” for item_moved. For all subsequent steps, the timing field contains the timestamps in milliseconds (ms) for when the participant started and finished the respective ranking step. The item_moved field records the Qualtrics item ID of the item that was dragged. The final field records the order of all items after completing that ranking step.

Take the participant with ID “ID93” shown in Figure 6. The initial entry for this participant reads {DROPT1|0|NA|4,3,5,1,2,6}, indicating that the initial randomized order was 4,3,5,1,2,6. Participant ID93’s second entry in “DROPT_RankProcess” reads as follows: {DROPT1|595,944|3|4,5,1,3,2,6}. This means that the participant started this ranking step after 595 ms and finished after 944 ms. The order *after* completing this ranking step was 4,5,1,3,2,6. Because the item_moved field is 3, we know that the participant moved item 3 from position two to position four.

When two neighboring items switch positions, the rankings per step alone do not reveal which item was moved. This is for instance the case for participant with ID “QX36”: When closely examining the initial ranking (5,1,2,4,6,3) and the ranking after step one (1,5,2,4,6,3), we cannot determine whether this participant moved item 5 *down* or item 1 *up*. In such cases, it becomes necessary to turn to the entry in item_moved. For participant QX36, this entry shows that item 5 was moved, allowing us to determine that item 5 was moved down rather than item 1 being moved up.

When analyzing the data, it can be useful to distinguish active moves from passive list shifts. An active move refers to the item that the participant directly drags and drops. A passive

shift refers to the automatic repositioning of other items when the dragged item is inserted above or below them in the list (see Figure 2). DROPT records both the resulting full list order after each step and the actively moved item. Thus, the ``current_order`` field captures the full list state after each move, whereas the ``item_moved`` field identifies the item that was actively dragged. In our analyses, we focus on active movements.

With this general description of the data in mind, we now turn to a five-step data processing and analysis script for deriving item-level measures of drag count, order, and distance. A template code to conduct these steps, “DROPT_5_Steps_Script.R”, can be found on the OSF.

Step 1: Data Formatting & Exclusions. The goal of this step is to create a clean data frame from the recorded data. The script first parses the cumulative “DROPT_RankProcess” log, separates the individual ranking steps, extracts the task ID, timing information, moved item, and current order, and checks whether each respondent $\times$ task log is valid. To ensure data quality, we generally exclude respondents with order respondent-task logs in an unexpected format, missing or invalid moved-item entries, or otherwise invalid logs. As with most paradata-based approaches, the DROPT procedure is susceptible to recording errors, most notably when the JavaScript used to capture drag-and-drop behavior does not function as intended (e.g., because JavaScript is disabled), which can result in missing or malformed logs (McClain et al., 2019). In Study 1B, this led to the exclusion of four respondents (4%) in both ranking tasks. However, researchers planning to apply DROPT in panel or longitudinal settings should account for the possibility that recording errors may compound across waves.

Step 2: Drag Count. This step creates a separate data frame, which records drag counts per-item and rank-order task in a long data format. Each participant will have exactly one

observation per item (in the case of Study 1B, 80 participants * 6 options = 480 observations in each task).

Step 3: Drag Order. This step is conceptually similar to step 2 and generates a long-format data frame with one column for the drag order. The item touched first will receive order “1”, etc. If an item is moved more than once, the drag-order measure is based on the first time the item was moved. The way untouched items are coded is up to the researcher. Our pre-registered approach in Study 1B was to assign the highest observed drag order plus one. That is, if only one item is touched, every other item will receive “2” as its drag order.

Step 4: Drag Distance. Step 4 analogously creates a long-format data frame in which the drag distance between two consecutive ranking steps is stored on an item level. The measure is based on the first observed movement of each item. The resulting dataset contains, for each respondent, rank-order task, and item, drag distance measures and the time taken to complete the drag-and-drop move. Drag distance captures how many items the moved element passes over before settling into its final position. The script returns both “distance” and “distance.r”. In “distance”, negative values indicate upward movement in the ranking and positive values indicate downward movement. In “distance.r”, this coding is reversed so that upward movement is positive. In the case of Study 1B, with six rank-order options, these measures range from -5 to 5. To deal with undragged items, our pre-registered approach was to assign a drag distance of zero.

Step 5: Merging Data Frames. In the final step, we merge the outputs from Steps 2 to 4 into a single, long-format dataset. This table combines, for each respondent × rank-order task × item, the per-item drag count, order, distance, and drag time. This is the final output, which can be used for further analysis.

Researcher Degrees of Freedom. We recommend that researchers decide and pre-register their analytic strategy beforehand. For *drag count*, responses can be recorded as binary or as the number of counts. For *drag order*, researchers must decide how to treat items that were either untouched or touched more than once. In our studies, we coded untouched items as the maximum observed order plus one and only considered the first touch. We used this approach because it preserves the ordinal meaning of the touch sequence and avoids losing data by retaining information about untouched items. However, alternatives for untouched items exist, for instance, dropping these observations or coding them as the maximum number of items. Dropping untouched items might be appropriate when only *active* choices are considered, or when there is a very small number of untouched items. For *drag distance*, similar decisions apply. We considered only the first drag-and-drop per item and coded the distance of untouched items as zero (instead of, for example, dropping these observations).

Compact R Function. In addition to the tutorial script, we provide a compact R function for easier usage: `DROPT_R_Function.R` on the OSF. After sourcing the function, users can call `process_drop_sequence()` on their Qualtrics data. The function implements the same five-step processing logic described above internally and returns a DROPT data frame.

The function is designed to be adaptable to different study designs. It can automatically process one or more rank-order questions in the same survey and does not assume a fixed number of answer options. By default, untouched items receive the highest observed drag order plus one (``untouched_order = "max_observed_plus_one"``), and malformed respondent-task logs are dropped while other valid tasks from the same respondent are retained (``malformed_action = "drop_task"``). These function parameters only need to be specified when researchers want to change them. For instance, users can instead set ``untouched_order = "max_possible"`` (to assign

the number of items in a rank-order task to untouched items) or ``untouched_order = "NA"` (to assign NA to untouched items), and they can set ``malformed_action = "error"` if they want the function to stop when malformed logs are detected. For example, the function can be used with the default settings as follows:

```
source("DROPT_R_Function.R")  
  
DROPT.df <- process_dropt_sequence(data)
```

Here, “data” refers to the data frame specified by the user. For the rank order variable, researchers can also explicitly change the coding of untouched items or the handling of malformed logs:

```
DROPT.df <- process_dropt_sequence(  
  data = data,  
  untouched_order = "max_possible",  
  malformed_action = "error"  
)
```

For diagnostics, users can set ``return_intermediate = TRUE``. This returns the final ``DROPT.df`` as well as intermediate objects, including the parsed ranking process, task metadata, separate drag-count/order/distance data frames, malformed entries, and dropped respondent-task logs. These outputs allow researchers to inspect recording problems, check item mapping when answer options were randomized, and document exclusions. For example:

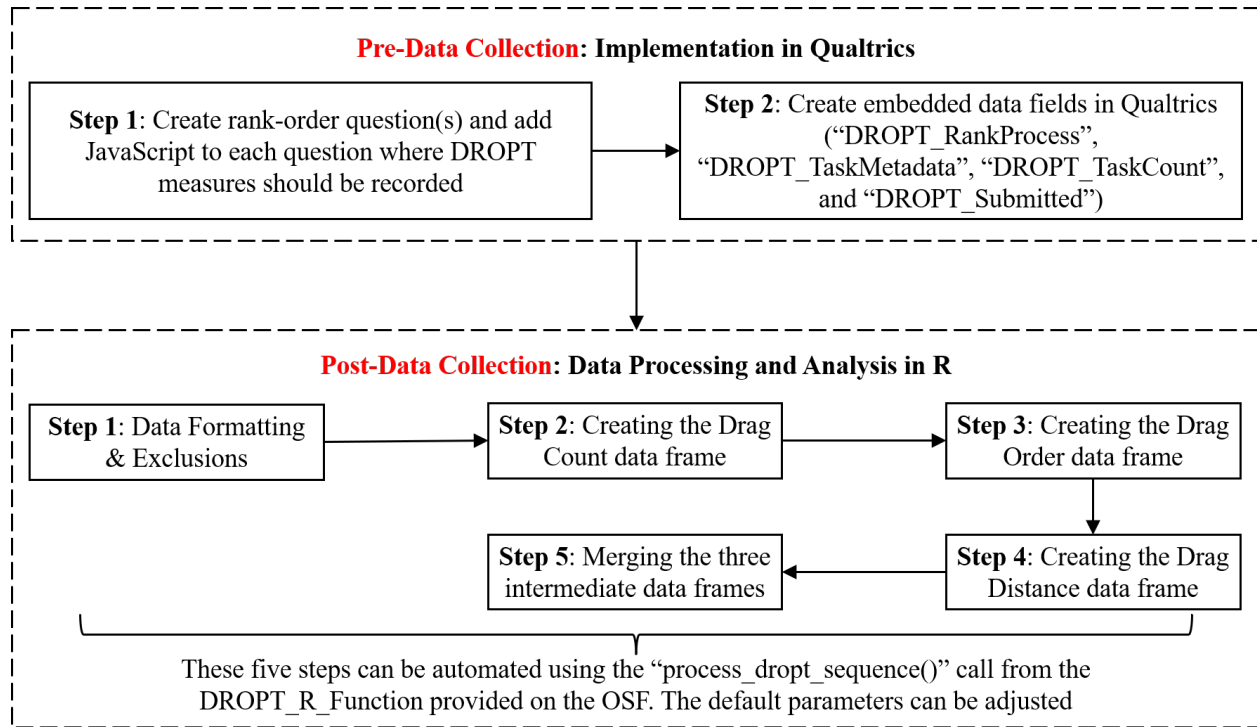
```
dropt_objects <- process_dropt_sequence(  
  data = data,  
  return_intermediate = TRUE  
)
```

Users can then extract the final dataset and the diagnostic data frames from this list:

```
# Final DROPT dataset  
DROPT.df <- dropt_objects$DROPT.df  
  
# Individual malformed log entries and the reason each failed validation  
MalformedEntries.df <- dropt_objects$MalformedEntries.df  
  
# Summary of respondent-task logs dropped because they contained malformed entries  
DroppedTasks.df <- dropt_objects$DroppedTasks.df
```

Implementation Checklist and Further Considerations

Implementation Checklist. Figure 7 shows a summary of the pre- and post-data collection steps involved to implement DROPT. As can be seen from the flowchart, the post-data collection steps can be automated using the DROPT_R_Function, which is provided on the OSF for this paper.

Figure 7*Flowchart of Steps to implement DROPT*

Implementation Reliability and Device Compatibility. Because DROPT relies on JavaScript, researchers should test the implementation before data collection in the same survey environment in which the study will be run. The JavaScript records both mouse- and touchscreen-based interactions by listening for mouse and touch events. Thus, the procedure can be used on desktop devices as well as touchscreen devices, provided that the Qualtrics rank-order question renders correctly and JavaScript is enabled. If JavaScript is disabled, blocked by a browser setting, or fails to execute, the DROPT embedded data fields will not be populated as expected, resulting in missing or malformed logs. These cases can be identified during the processing steps in R using the intermediate data set.

Empirical Results of Study 1B

The final sample included 80 respondents in both tasks. Respondents demonstrated high engagement across tasks, making on average four ranking adjustments ($M_{\text{color}} = 4.04$, $SD = 1.23$; $M_{\text{corner}} = 4.01$, $SD = 1.25$) and completing the tasks accurately ($M_{\tau[\text{color}]} = 1.00$, $SD = 0.00$; $M_{\tau[\text{corner}]} = 0.98$, $SD = 0.13$).

Ranking patterns aligned closely with the results of Study 1A. As Figure SM4 shows, under reversed ranking instructions, we observe that shapes with lighter colors and fewer corners are moved up earlier and dragged further distances. Similarly, regression results support the 3F interpretation, with all relevant predictors significant in the hypothesized direction (see bottom panel of Figure 3). As before, we also calculated correlations between the DROPT measures after residualization. We found moderate to high correlations in both the color ($M_{\text{count-order}} = -0.76$, $SD = 0.17$; $M_{\text{count-distance}} = 0.42$, $SD = 0.43$; $M_{\text{order-distance}} = -0.59$, $SD = 0.48$) and the corner tasks ($M_{\text{count-order}} = -0.71$, $SD = 0.21$; $M_{\text{count-distance}} = 0.32$, $SD = 0.44$; $M_{\text{order-distance}} = -0.51$, $SD = 0.51$). Given the relatively high correlations, we focus our analysis on one measure, drag distance, in Study 2.

Discussion

When participants are instructed to rank-order items by an objective, perceptual attribute, they follow a sorting approach that aligns with a selection sort pattern. Accordingly, we find a 3F pattern consistently across studies. Participants dragged items scoring higher in the relevant dimensions more *frequently*, *first*, and *further* up. Under the assumption that the item being moved is the option currently receiving consideration, this pattern suggests that participants identify the highest-valued remaining item, move it toward the top of the list, and then move on to the next item. Study 1B revealed that the 3F pattern is robust to reversing instruction and that

ranking behaviors, on average, most closely align with the proposed selection sort rather than predictions derived from the markedness literature.

We also observed that, at times, irrelevant attributes emerged as significant predictors of drag count, order, and distance, although their effects were substantially weaker in magnitude. A possible explanation for instances where irrelevant attributes predicted ranking behavior is that participants may have begun with a selection sort strategy but switched strategies later or applied the strategy inconsistently.

Overall, study 1 establishes how participants behave when ranking by objectively known attributes. Although we do not claim that DROPT is a direct measure of attention, these findings suggest that DROPT process measures may serve as potential indicators of attention and attribute use during ranking. That is, when participants rank by a given attribute, they appear to adopt selection sort and selectively consider items that score highly on that attribute and move them upward in the list. This allows us to address further questions, for instance, which attributes do people use when instructed to rank by subjective attributes? In Study 1, we explicitly asked participants to focus on only one of the two attributes in our tasks. But how do people weigh attributes without instructions, such as in a preference task where options are characterized by multiple attributes? We investigate this question in Study 2.

Study 2: Ranking Lotteries

In Study 2, we illustrate one application of DROPT by examining how ranking behavior can potentially inform our understanding of the preference construction process in a multi-attribute setting.

To do this, we asked respondents to rank lotteries (also referred to as bets) to reflect their personal preference. Each lottery was described by a probability of winning and a payoff (see

Figure 1B). None of the lotteries dominated the others. That is, none of the lotteries was better (or worse) than the others on both attributes, and all had similar expected values. This was intended to encourage tradeoffs between these two attributes. As an inevitable byproduct, however, this resulted in a high correlation between the attributes. We address the implications of this correlation in the Results section. Study 2 aims to address three key research questions:

1) *Does the 3F pattern observed in Studies 1A and 1B with shape stimuli generalize to lotteries characterized by probability of winning and payoff?*

2) *Is there a relationship between the drag-and-drop process data and individuals' preferences?*

3) *Can drag-and-drop process data in the preference task improve predictions of preferences in subsequent, independent choices?*

To address the first question, we designed two initial blocks in which we asked respondents to rank the lotteries from highest to lowest probability and from highest to lowest payoff. These *attribute ranking tasks* are conceptually equivalent to the tasks used in Study 1 (see *Study 2 Supplementary Methods* for stimuli details).

To address the second question, we examined data from a third block—the preference ranking task—in which respondents ranked all lotteries in order of preference. Our interest was in whether participants' drag-and-drop behavior during the ranking process was associated with their preferences. For example, participants who prefer higher-payoff lotteries may tend to drag and drop higher-payoff options more frequently, earlier, and farther upward in the ranking.

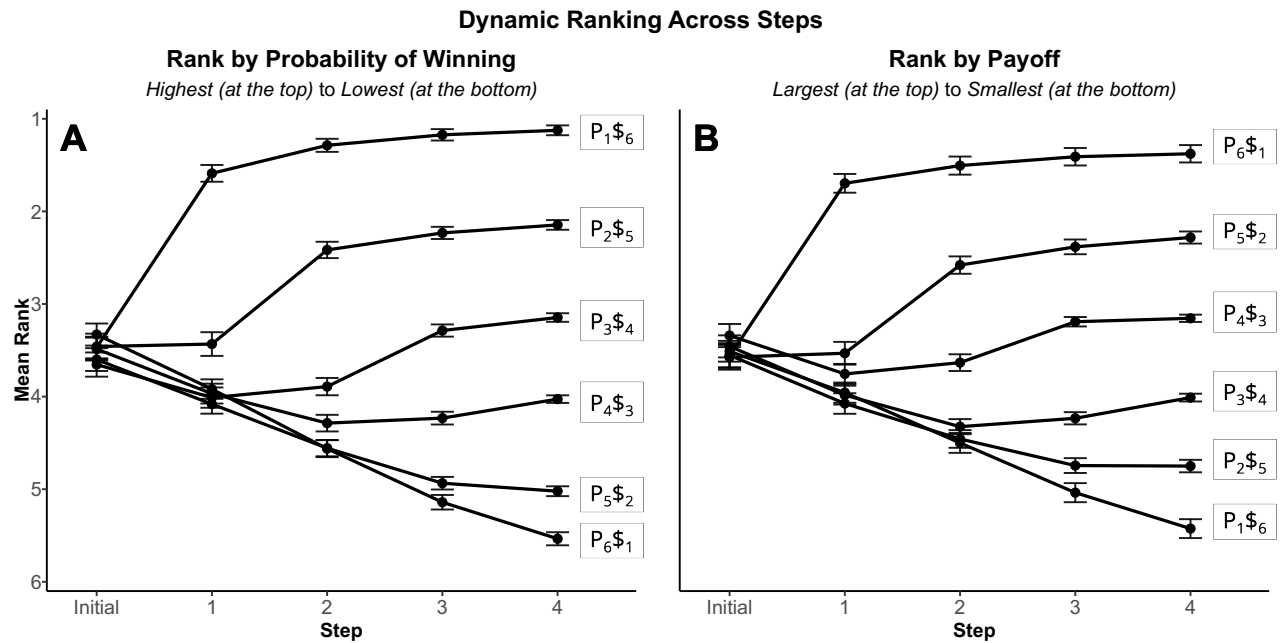
Finally, to address the third question, we used the process data from the preference ranking task to predict choices in two subsequent validation tasks. These included a *validation ranking task*, where participants ranked a new set of lotteries by preference, and a *validation*

choice task, which involved five binary lottery choices. We hypothesized that process data can capture additional elements of preference beyond those reflected in final rank data alone, thereby improving the prediction of behavior in these subsequent tasks.

In summary, we examined tasks involving subjective preferences in a multi-attribute setting, where decision makers were likely to integrate different attributes to form preferences.

Results

We begin by examining whether the process behavior from the perceptual tasks in Study 1 replicates when participants rank lotteries by one single attribute (see *Study 2 Supplementary Results* for descriptive statistics). For both the probability and the payoff tasks (see Figure 8), the observed ranking patterns were consistent with those for the perceptual tasks and conformed to the selection sort strategy: The highest scoring option was first moved to the top position (from a random starting location), followed by the remaining lotteries in order of their value of the relevant attribute. Were individuals relying on probability or payoff to complete the strategy? We replicated the 3F regression analyses performed in Study 1. However, as noted, the attributes in the current stimuli were highly correlated ($r = -0.71$), which may limit our ability to disentangle attribute usage through a modeling approach. Regression results showed that the task-relevant attribute significantly predicted drag-and-drop behavior, consistent with expected attribute usage. However, the task-irrelevant attribute also emerged as a significant predictor (see statistical details in *Study 2 Supplementary Results*). We believe this pattern more likely reflects the high correlation between attributes introduced by the stimulus design and the resulting modeling constraints, rather than meaningful use of the task-irrelevant attribute. We discuss this issue further in the SM.

Figure 8*Steps of Ranking Lotteries by Probability of Winning (A) and Payoff (B)*

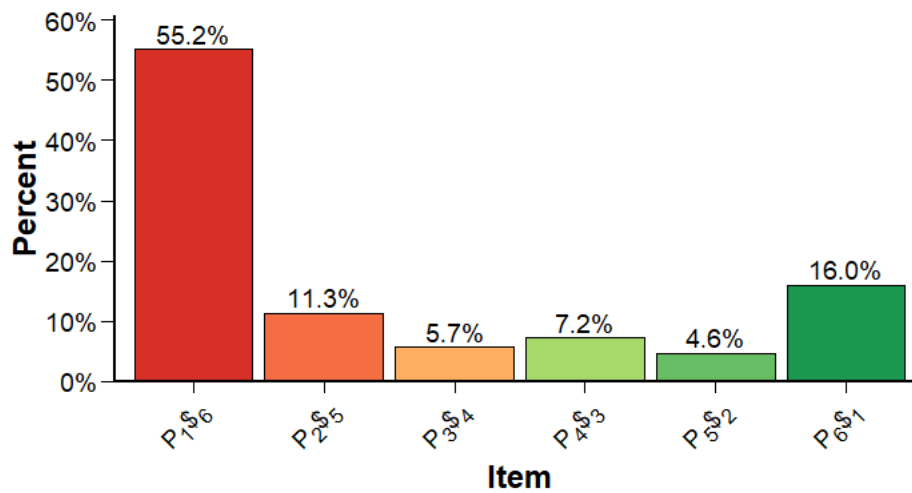
Note. Mean rank of each lottery across drag-and-drop steps in probability of winning (Panel A) and payoff (Panel B) tasks. Each line represents one lottery. Error bars represent ± 1 standard error across respondents at each step. The first four drag-and-drop steps are displayed. Mean rank changes reflect both active movements and passive rank shifts of items.

We next turn to our second question: Is there a relationship between process data and individuals' preferences? We focus on the preference ranking task, where participants ranked lotteries according to their personal preference. First, we found substantial heterogeneity in preferences. Figure 9 shows how often each of the six lotteries was ranked as the most preferred. The bar colors represent the probability of winning, ranging from red (the option with the highest probability of winning) to green (the option with the highest potential payoff). Slightly more than half of participants (55.2%) ranked the lottery with the highest probability of winning as their top choice. The remaining participants ranked other lotteries first, most commonly the option with

the highest payoff. To analyze these differences, we categorized participants into two primary profiles: The P-bet group: Participants who ranked the highest-probability lottery first. The \$-bet group: Participants who ranked the highest-payoff lottery first. The P-bet and \$-bet groups correspond to the leftmost and rightmost bars in Figure 9, respectively.

Figure 9

Distribution of the Most Preferred Lottery in the Preference Ranking Task



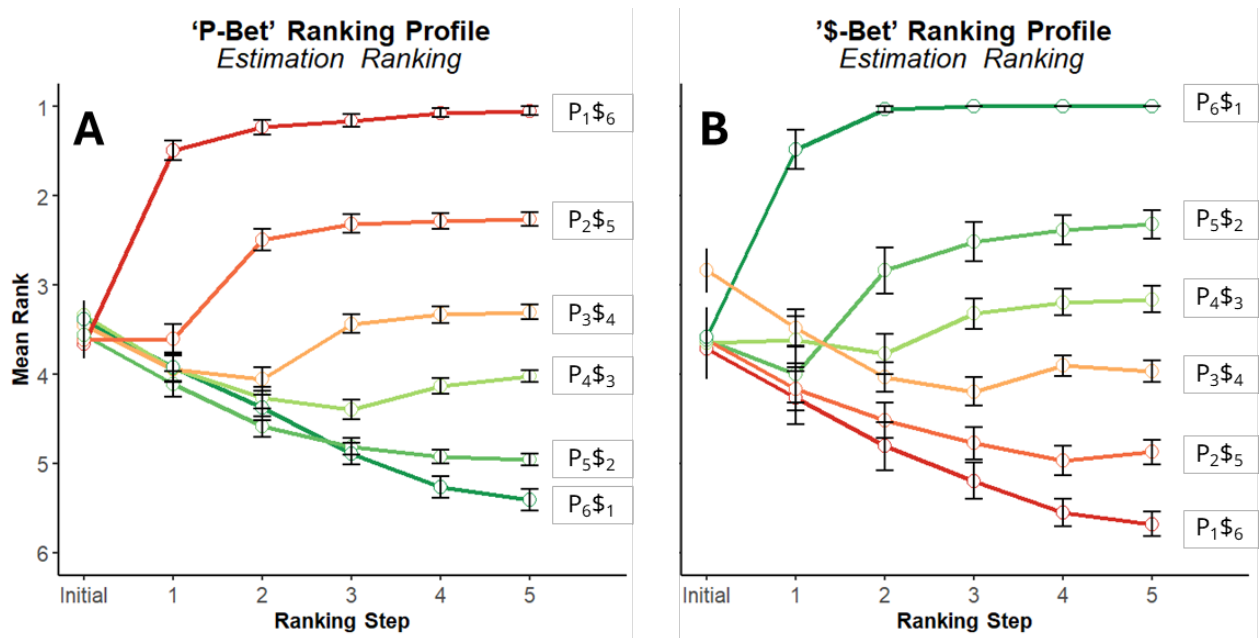
Note. $P_1\$6$ denotes the lottery with the highest probability and the lowest payoff, whereas $P_6\$1$ denotes the lottery with the lowest probability and the highest payoff. For example, the leftmost bar shows that 55.2% of participants ranked the highest-probability lottery first.

We examined the sequence of drag-and-drop actions for these two groups to describe how their preference rankings unfolded. Several ranking patterns were possible. Participants might compare adjacent items and swap them incrementally; alternatively, they may identify and demote the less attractive options (e.g., pushing lowest probability bet to the bottom). Conversely, they can identify and promote the more attractive ones and rank them toward the top.

Figure 10 shows the drag-and-drop behavior of the two groups, which suggests that individuals tended to construct preferences by promoting options they liked most, moving them more frequently, earlier, and farther upward. For the P-bet group, lotteries with higher probabilities were moved early in the ranking process; for the \$-bet group, the reverse was true: The higher-payoff lotteries were moved first. Given the link between attribute usage and the 3F pattern, these results were consistent with the idea that P-bet group primarily relied on probability when ranking lotteries by preference, and \$-bet group on payoff.

Figure 10

Ranking Behavior of P-Bet and \$-Bet Groups in the Preference Ranking Task



Note. Error bars represent ± 1 standard error across respondents at each step. The first five drag-and-drop steps are displayed for each respondent. Mean rank changes reflect both active movements and passive rank shifts of items.

These observational patterns were consistent with the selection-sort pattern observed in the perceptual tasks and suggest an association between preference and the process data, which

may provide useful preference-relevant information for modeling choice behavior. Here, we explore two aspects of the preference-assembling process that DROPT data may inform: First, while it is challenging to identify the weights each individual places on probability and payoff given the relatively sparse choice data revealed by a single ranking task, drag-and-drop patterns provide additional information that can help recover such weights more precisely. This role motivates a model that estimates participant-specific attribute weights using DROPT measures, which are then used to modulate the influence of probability and payoff when predicting choices. Second, drag-and-drop movements may help understand attention allocation to options during ranking. This role motivates a model in which drag distance is entered directly as an additional process-based predictor, capturing option-level process information that may not be fully reflected in final ranks alone. We use drag distance as a compact summary of participants' ranking process. This choice was motivated by the fact that, under a selection sort ranking process, the three process measures converged and drag distance may provide especially fine-grained information by capturing both the direction and magnitude of each drag-and-drop movement.⁴

Exploring both two approaches allows us to ask whether DROPT data improve prediction by refining estimates of individual attribute weights, by capturing additional process information during the prediction task, or both. Notably, the first approach may be especially useful because it tests whether process data from one ranking task can capture individual differences that improve prediction in subsequent tasks, without requiring new process data at the time of prediction.

⁴ We do not suggest that drag distance is generally the most informative process measure. As reported in the simulations, when participants use other sorting strategies, these measures may be less overlapping, and drag count, drag order, or other process measures may be equally or more informative.

We examine whether incorporating process data improves the in-sample fit of choice in the preference ranking task and the prediction of choices in subsequent tasks. We compared four models that differed in whether and how they incorporated process data. We begin with a *baseline* model in which probability and payoff, the two attributes defining each lottery, are included as predictors. This first model assumes homogeneous weighting of probability and payoff across participants. A second, *individual random-effects* model, allowed these attribute weights to vary by participant. These first two models represent the standard case in which only final rankings are observed. A third, *distance-weight augmented* model, augmented the random-effects specification by incorporating each participant's drag distances from the preference task as *process-based weights* on probability and payoff (see *Study 2 Supplementary Methods, Preference Ranking Task*, for model specification details). In other words, this model differs from the individual random-effects model in that participants' weights on probability and payoff are based not only on their final rankings, but also on drag distances during the ranking process. Finally, a fourth, *distance augmented model*, takes a different approach: rather than using process measures to inform attribute weights, it keeps the individual random-effects specification intact and adds drag distance directly as an additional predictor. This is the only model among the four that requires process data from the new prediction task when making predictions. In these two process models, we adopted a residualized drag distance measure by regressing raw drag distance on each item's initial position while including participant and item random intercepts, thereby partialing out variance attributable to initial position and participant- and item-level differences. Conceptually, the third model used drag distance to *modulate individual attribute*

weights estimated from final rankings, and the fourth model treated drag distance as an additional process-based variable that may capture attention to options.⁵

A potential limitation of these models is that probability and payoff were negatively correlated, and analyses of the prior ranking tasks by objective attributes suggest that correlations make it difficult to estimate attribute weights. To explore this, we conducted a simulation to provide a sensitivity analysis focused on the individual random-effects model. Results suggest that while the correlated design may limit the precision of estimates, it does not necessarily preclude recovery of meaningful preference weights (see the *Sensitivity Analysis* section of the SMs for details; the simulation code is available on OSF). Finally, the correlation affects all four models and therefore should not undermine comparisons of relative model fit or the conclusion that incorporating process data captures additional preference-relevant information.

We first estimated the four models on the preference ranking task. Model comparisons based on log-likelihood, AIC, and BIC in Table 2 show that the individual-level model fits better than the baseline model, and that incorporating drag distance process data further leads to substantial improvements in model fit. These results suggest that the process data may contain preference-relevant information beyond the final ranking alone.

⁵ For the distance-weight augmented model, one alternative is to use raw drag distance while controlling for initial position and respondent random effects, instead of residualized drag distance.

Table 2*Comparison of Model Fit Indices in the Preference Ranking Task*

Fit Indices	Baseline	Individual Random-Effects	Distance-Weight Augmented	Distance Augmented
-LL	1845.57	1470.67	1279.97	1279.73
AIC	3709.13	2969.34	2595.93	2589.59
BIC	3754.09	3039.28	2685.86	2664.52

Note. The “Distance Augmented” model is the only one that will require drag-distance data from the preference ranking task when generating predictions for the validation ranking task.

To assess if the information captured by the process data is helpful in predicting choices in subsequent, independent tasks, we turn to the two validation tasks. In the validation ranking task, participants ranked a new set of six lotteries. As in the preference task, these lotteries had approximately equal expected values but differed in probability and payoff, with small random jitters added so that each participant saw slightly different values (see Table SM5).

For each participant, we used the model coefficients estimated from the preference task to compute a latent preference score for each of the six new lotteries in the validation ranking task. We assessed predictive accuracy with two metrics. First, for each participant, we assigned predicted ranks based on the latent utilities and compared them to observed rankings using Kendall’s τ . Second, we computed the Pearson’s r correlation between the predicted latent utilities and observed final ranks. Although Pearson’s r is typically used for continuous variables, treating ranks as approximately continuous allows this measure to capture whether relative differences in latent preferences align with the observed rank, beyond the ordinal relationship captured by τ . As shown in Table 3, the *individual random effects* model outperformed the *baseline* model, and notably, including drag distance measures yielded

additional gains in predictive accuracy compared to the individual random effects model.

Exploratory analyses suggest that these accuracy improvements were statistically significant for the Distance Augmented model, and the corresponding evidence for the Distance-Weight Augmented model was weaker (see *Study 2 Supplementary Results, Validation Ranking Task*, for statistical details).

Additionally, we examined whether the predictive value of drag distance depended on participants' preference patterns. In particular, process data may be especially useful when final rankings provide a weaker or more ambiguous signal of participants' preferences. Participants were classified as having mixed preference patterns when their lottery rankings were not strictly ordered by either probability or payoff. Incorporating drag distance produced larger predictive gains for these participants than for those whose rankings followed a single attribute strictly. Among participants with strictly ordered rankings, adding drag distance did not improve out-of-sample prediction.

Table 3

Comparison of Prediction Accuracies in the Validation Ranking Task

Accuracy Metrics	Baseline	Individual Random Effects	Distance-Weight Augmented	Distance Augmented
Kendall's τ	$M = 0.38$ [0.27, 0.48]	$M = 0.56$ [0.48, 0.64]	$M = 0.58$ [0.50, 0.66]	$M = 0.75$ [0.70, 0.80]
Pearson's r	$M = 0.40$ [0.30, 0.51]	$M = 0.59$ [0.51, 0.68]	$M = 0.61$ [0.53, 0.69]	$M = 0.80$ [0.75, 0.85]

Note. Pearson's r reflects the correlation between each participant's predicted latent utility scores and their observed final rankings. Kendall's τ reflects the correlation between predicted and observed rank orders. The table reports the mean and the 95% confidence interval of each prediction accuracy metric ($N = 182$). See the *Study 2 Supplementary Results, Validation Ranking Task* for exploratory inferential tests comparing these models.

Finally, the validation choice task presented five binary choices between a higher probability, lower payoff option (a P-Bet) and a lower probability, higher payoff option (a \$-Bet). Because this task involved a different response mode than the ranking task, we estimated a different set of models. To test whether process data improved predictive power, we compared logistic regression models that differed in the information carried over from the preference ranking task. A *baseline* model included only differences in probability and payoff in each pair of lottery choices. A second, *final-rank-weight augmented* model, augmented the *baseline* model by weighting each attribute difference using participant-specific weights on probability and payoff estimated from the preference task choice data. The rationale behind this approach is that individual differences in weighing probability and payoff, as revealed in the preference task, should help improve predictions of choices. A third, *final-rank and distance weights augmented* model, applied the same weighting logic and further augmented weights derived from participants' residualized drag distance in the preference task (see Table SM10 for specification details and additional benchmark models). We can assess whether process-based weights improve predictive accuracy above and beyond choice-based weights alone by comparing the second and the third model.

These models were evaluated using 80/20 cross-validation across participants ($N = 182$), with the procedure repeated 5,000 times to ensure a reliable mean performance estimate across different possible sample splits. The *baseline* model achieved a mean prediction accuracy of 66.03% in the held-out data across random sample splits ($SD = 5.13\%$), and adding final-ranking weights increased the mean accuracy to 74.41% ($SD = 4.35\%$). The third model that incorporated drag distance information outperformed the second model in 70.70% of the 5,000 iterations and yielded a further, though modest, improvement in mean accuracy to 74.98% ($SD = 4.30\%$).

Consistent with results from the validation ranking task, gains from incorporating the residualized drag distance weight were more pronounced among participants with more mixed preference (see *Study 2 Supplementary Results, Validation Choice Task* for these results).

Discussion

In Study 2, we replicated with lottery stimuli the drag-and-drop pattern: When participants ranked lotteries by probability of winning and payoffs, they moved items higher in the task-relevant attribute more frequently, earlier, and farther upward, consistent with a sequential upward ranking strategy. We further found that this pattern aligned with participants' preferences: those who selected the highest-probability lottery as their top choice showed greater reliance on probability during ranking, whereas those who selected the highest-payoff lottery showed the corresponding pattern for payoff. These results suggest that, when ranking by preference, participants may similarly construct their rankings by sequentially considering which option best satisfies their preferred attribute and moving that option upward.

Finally, we showed that incorporating the drag distance measure improved predictions of participants' preference in subsequent, independent validation ranking and choice tasks. These improvements were more pronounced among participants with more mixed preferences, suggesting that process data may be most informative when final rankings alone leave greater ambiguity about preferences. A potential boundary condition, however, is that process data may provide less additional value when final rankings already provide a strong signal of participants' preferences. These results suggest that DROPT can potentially record movement patterns that are meaningfully related to preference construction in this task.

General Discussion

Individuals routinely make ranking decisions, from choosing universities or restaurants to prioritizing personal goals, and these decisions often involve complex tradeoffs between competing attributes. In the present research, we introduced and validated a new process-tracing paradigm, Dynamic Rank-Order Process Tracing (DROPT), which records how participants move options while constructing rankings in a drag-and-drop interface. Under the assumption that the item being dragged is likely to reflect the option currently being considered, these interaction behaviors can serve as potential indicators of the unfolding judgment and preference-construction process.

Across three studies, we found consistent evidence that participants moved items higher in the task-relevant attribute more frequently, earlier, and farther upward. This 3F pattern replicated across perceptual attributes (e.g., number of corners, color darkness) as well as lotteries varying in probability and payoff. Benchmark simulations further showed that this pattern was most consistent with a selection-sort process, in which participants iteratively identify the highest-value remaining item and move it into position, rather than being a necessary consequence of the interface or final ranking constraints.

Furthermore, these drag-and-drop behaviors could reflect participants' underlying preferences, as individuals who favored high-probability versus high-payoff lotteries showed systematically different movement patterns. Participants who ranked high-probability lotteries at the top moved these lotteries more frequently, first, and further upward, consistent with greater reliance on probability during ranking. Those who ranked the high-payoff lotteries at the top showed the corresponding pattern for payoff.

Beyond describing process-level patterns, we demonstrated with the drag distance measure that DROPT data can enhance predictive accuracy in subsequent, independent choices,

suggesting that DROPT may contain preference-relevant information that extends beyond final rankings. Although the gains in predictive accuracy are modest—particularly for the distance-weighted augmented model in the validation ranking and validation choice tasks—these small improvements may be important at scale, where such gains can accumulate across many predictions. In addition, these improvements were more pronounced among participants with more mixed preferences, that is, participants who did not have a dominant preference for a high probability or payoff in Study 2. For these participants, preferences are presumably harder to infer from final rankings alone. This pattern also points to a potential boundary condition: when preferences are strong and final choices already provide sufficient information, incorporating process data may offer limited additional benefits.

Prior research has examined how rank information influences decisions (Chun & Larrick, 2022; Weingarten et al., 2023). Most notably, Chun and Larrick (2022) showed that individuals disproportionately attend to top-ranked options, over and above the underlying attribute information. However, much less research has examined the *process by which rankings are constructed*. DROPT addresses this gap by recording how participants move items while forming rankings in a drag-and-drop interface. These movement data provide a simple way to examine how participants rely on attributes during rank construction. This makes DROPT potentially very useful in multi-attribute settings where researchers traditionally only observe final rankings, limiting their ability to infer how different attributes were used during the decision process.

DROPT also contributes to the limited literature on using paradata to explore substantive variables. In our study, we show that JavaScript-based paradata can support inferences about ranking behavior and preference construction in drag-and-drop ranking tasks. This goes beyond common applications of paradata, which have often focused on providing information into

response quality and errors associated with the survey data collection process (McClain et al., 2019; Vehovar et al., 2024).

As with any research, the current work has limitations and points to several promising directions for future research. First, our cognitive interpretation of DROPT depends on the assumption that the item being dragged and dropped reflects the option currently under active consideration. This assumption is consistent with prior process-tracing research, which treats temporally adjacent information-acquisition behaviors as indicators of the information being considered (Costa-Gomes et al., 2001). Nevertheless, we caution that it may be premature to treat DROPT as a direct measure of attention. Future research could more directly test this assumption by combining DROPT with complementary process measures, such as eye-tracking or verbal reports.

Another limitation of our empirical design is the high correlation between the two attributes used in Study 2: Probability and payoff. In designing our tasks, we aimed to minimize differences in expected value to avoid making any one lottery inherently more attractive than another. However, this approach inevitably leads to a high correlation between probability and payoff. Although our sensitivity analysis suggests that the correlated design does not preclude meaningful recovery of preference weights, it may be relatively simplified and cannot substitute for a fully orthogonal design. We encourage future research to use orthogonal multi-attribute designs to better isolate the individual effects of each attribute, similar to the near-zero correlation in Study 1. In addition, our prediction analyses used a two-stage modeling approach. Specifically, participant-level coefficients were first estimated from the ranking model and then used as predictors in subsequent choice-prediction models. This approach is practical, but it does not fully propagate uncertainty from the first-stage estimates into the second-stage analyses.

Thus, these results should be interpreted as an initial predictive demonstration. Future research could address this limitation by estimating the ranking and choice components in a single joint model, for example in a hierarchical Bayesian framework.

A further limitation of our empirical design is that we only considered lists of six items. Specifically, participants rank-ordered six shapes in Studies 1A and 1B and six lotteries in Study 2. With this design, we observe a 3F pattern for attributes. However, when ranking longer lists (e.g., 20 items), alternative sorting strategies may be preferred by decision-makers. For instance, participants may first place items with the highest *or* lowest values on a given attribute into their final positions, before refining the remaining ranks. Under such conditions, a sorting pattern based on attribute “extremeness” may emerge. Although we do not test this possibility in the present paper, DROPT is well suited to detect and characterize such alternative ranking processes. More broadly, because DROPT relies on drag-and-drop rank-order questions, the response format itself may shape participants’ behavior.

Several other promising directions for future research exist. In our studies, we used attributes known to researchers a priori: Number of corners, probability, etc. However, given the emergent 3F pattern we observed in our studies, could DROPT be used to *identify* relevant attributes? For instance, by analyzing the order of item movements, future work could use DROPT to understand which items participants attend to first or drag the largest distances. This, in turn, could offer insights relevant to understanding which attributes are relevant. It might be, for instance, that participants attend to irrelevant attributes, which might distort their judgments; an idea related to the Attribute Substitution framework (Johnson et al., 2024; Kahneman & Frederick, 2002; Ludwig et al., 2025). Recent work illustrates this promise. In a study by Sugerman et al. (2025), participants ranked a range of everyday products according to their

carbon emissions. Importantly, participants were not provided with actual emissions data; instead, the rankings reflected their own best estimates, and the authors experimentally manipulated which attribute participants were encouraged to substitute when making these emission judgments. The authors used DROPT data to show that the manipulated attribute predicted participants' drag-and-drop patterns only in conditions in which it was encouraged to be substituted, providing process evidence of attribute substitution.

Future work could also examine timing variables recorded by DROPT. In the present studies, we focused on drag count, order, and distance because these measures seemed most directly related to sorting strategies. However, DROPT also records timing measures, such as when participants begin moving items and how much time elapses between ranking steps. These measures are included in the DROPT output and are available for analysis. While this was not the main focus of this work, we encourage future studies to investigate how such timing measures recorded by DROPT could provide additional insight into ranking processes. Examining timing may also motivate richer approaches to modeling DROPT process data and incorporating it into choice prediction.

For ease of comparison, we held task complexity constant across studies, with six items and two attributes. However, this design constrains the possible range of trade-offs in the preference ranking tasks, where the majority of participants relied predominantly on either probability or amount. Under this relatively simple setting, incorporating DROPT variables improved prediction accuracy in subsequent, independent tasks, but the magnitude of this improvement was modest. We encourage future research to systematically vary the complexity of the ranking task, for instance, by increasing the number of items or attributes included. Varying this complexity may influence both the cognitive strategies participants use and the

ranking patterns observed during the task, including how individuals manage trade-offs among attributes. According to the adaptive decision maker framework (Payne et al., 1993) individuals simplify complex tasks by selectively focusing on a subset of attributes. This suggests that more complex ranking tasks may prompt participants to simplify the ranking task and focus on a smaller subset of items or attributes and trade-off between them. DROPT could dynamically capture such adaptive processes.

Future research could do so by examining the utility of DROPT in other multi-attribute domains, such as consumer product choice, health decisions, or policy preferences, where preferences may involve richer trade-offs and process data may be even more informative. Such work would also help evaluate the robustness of the observed accuracy improvements, examine whether the benefits of process-based measures depend on choice complexity and domain, and further clarify the mechanisms underlying these accuracy improvements.

As consumers, we encounter rankings in many different forms. As researchers, we often ask participants to produce rankings to address our research questions. In this paper, we introduced a new method that allows us to observe how ranking processes unfold over time. Process-tracing methods have long shed light on the psychological mechanisms that underlie decision-making. We hope that DROPT, as a new process-tracing approach, will further enhance our ability to study rank-order judgments and reveal mechanisms that conventional self-reports or outcome-based measures may overlook.

Declarations

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Conflicts of interest

The authors certify that they have no conflicts of interest to declare.

Ethics approval

This research was approved by the Columbia University Institutional Review Board (IRB Protocol #AAAB1301).

Consent to participate

Across experiments, participants provided informed consent prior to beginning the study.

Consent for publication

Participants provided informed consent prior to beginning the study, which indicated that their anonymized data may be used for research publications.

Availability of data and materials

See the Open Practices Statement below.

Code availability

See the Open Practices Statement below.

Authors' contributions

Xuwen Hua and Arian Trieb contributed equally to this manuscript and share first authorship. All authors contributed to the conception and design of the project. Xuwen Hua and

Arian Trieb collected the data, conducted the analyses, and drafted the manuscript. All authors contributed to revisions. Eric J. Johnson and Jonas Ludwig supervised the project.

Open Practices

The tutorial materials, data, analysis scripts and pre-registrations are available on the Open Science Framework (OSF;

https://osf.io/vaycw/overview?view_only=f7fe74ff872546cf8fc6841c94ed93c4). All studies were approved by Columbia Institutional Review Board (IRB Protocol #AAAB1301). We conducted all analyses in R, version 4.5.1 (R Core Team, 2025), using primarily the dplyr package (Wickham et al., 2025) for data manipulation and the ggplot2 package (Wickham, 2016) for data visualization. We employed ChatGPT (Versions 4.0-5.2) to assist with refining our coding.

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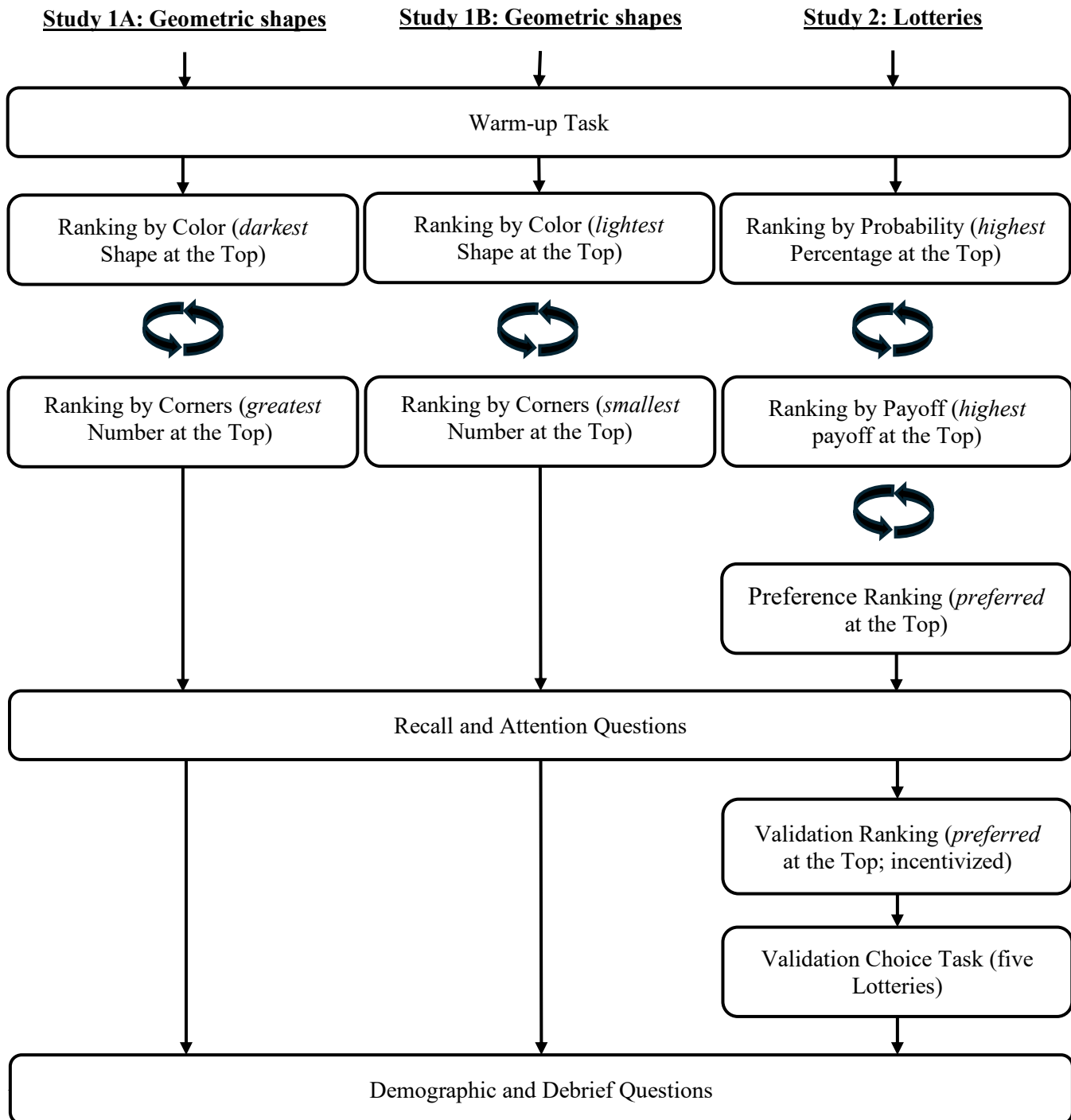
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Supplementary Materials (SMs)

Figure SM1

Conceptual Study Flow for Studies 1A, 1B, and 2.



Study 1 Supplementary Methods

Figure SM2 and SM3 show the interface participants saw in the rank by color and rank by corner tasks in Study 1A, respectively. The tasks in Study 1B were identical, except that the instruction specifying which items were ranked at the top versus the bottom was reversed; for conciseness, the interface is not shown here. See the OSF repository for the complete Qualtrics materials.

In Studies 1A and 1B, participants were asked to recall the ranking tasks they were asked to complete after finishing both tasks. This served to assess task comprehension and engagement. For instance, in Study 1A, participants were asked to select two options describing the tasks they were asked to complete. The response options were: (1) ranking shapes from darkest to lightest in color, (2) from most to fewest corners, (3) from simplest to most complex, (4) from most to least complex, and (5) from most to least symmetric. Results are reported in supplementary result sections below.

Across all studies, participants also completed a set of additional debrief questions at the end of the survey. These questions assessed the device participants used to complete the study (multiple choice: mouse [wired or wireless], trackpad [touchpad], touchscreen [finger or stylus], or other), whether participants encountered technical issues, whether they found the study confusing, whether they found the survey tedious (yes/no), and whether they used any external sources while completing the survey. When participants indicated that they experienced technical issues, the study was confusing or tedious, or that they have used an external source, they were prompted to provide an open-ended description of the issue or the external source they used. We report responses to these debrief questions below in the corresponding supplementary result sections to each study.

Across studies, we also collected open-ended responses in which participants were asked to describe what they believed the purpose of the study was. In Study 1A, participants additionally answered an open-ended question describing how they approached the attribute-ranking tasks. These open-ended responses were not analyzed in the current manuscript.

Figure SM2

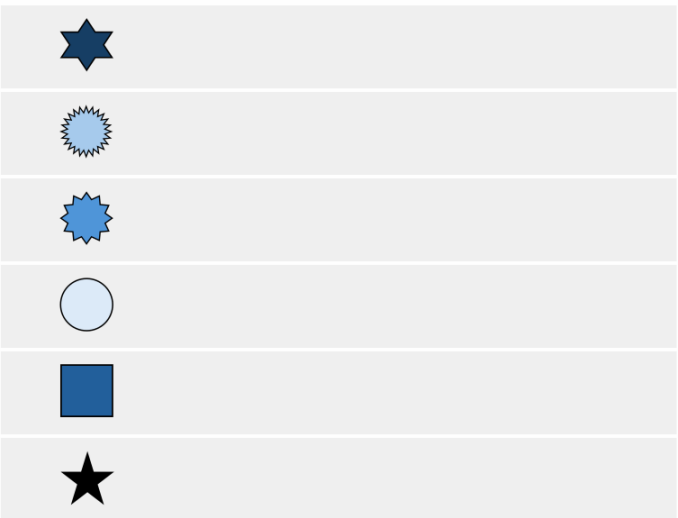
Task Interface for Rank by Color Task in Study 1A

Please rank the following shapes based on how dark in color they appear to you.

To rank them, click and drag each shape into order of the **darkest (at the top)** to the **lightest (at the bottom)**.

Remember, shapes with darker color should be placed **above** shapes with lighter color. Please try to ensure that every shape is in the right place.

Hint: For instance, most people find the color in  to be darker than the color in .










Note. The black outline is for visual clarity in the SM only and was not shown to participants.

Figure SM3*Task Interface for Rank by Corners Task in Study 1A*

Please rank the following shapes based on the number of corners they have.

To rank them, click and drag each shape into order from the **most corners (at the top)** to the **fewest corners (at the bottom)**.

Remember, shapes with more corners should be placed **above** shapes with fewer corners. Please try to ensure that every shape is in the right place.

Hint: For instance,  has more corners than .

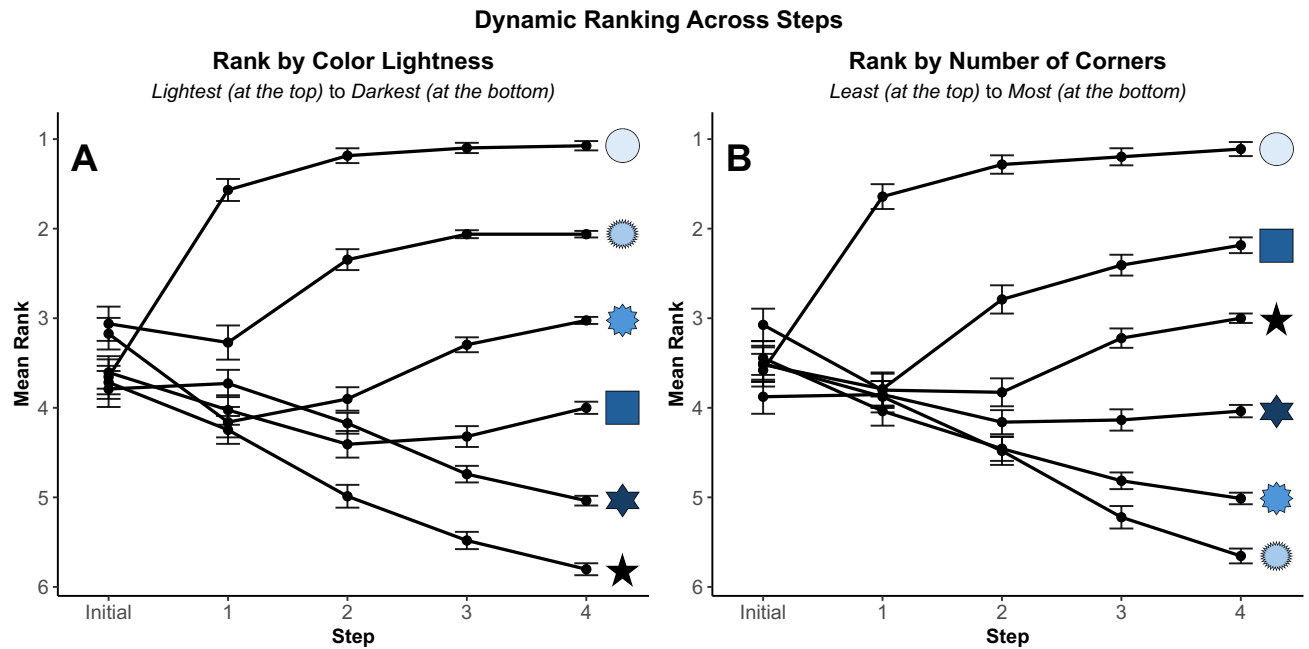
Note. The black outline is for visual clarity in the SM only and was not shown to participants.

Study 1 Supplementary Results

After initial exclusion, the median respondent education level was Bachelor's degree in college (4-year), and median income level was \$50,000-\$74,999 for both Studies 1A and 1B. For the recall questions, the majority of participants in both Study 1A (90.43%) and Study 1B (93.48%) answered correctly. We also summarize responses to the debrief questions for Studies 1A and 1B. In Study 1A, the majority of participants (65.96%) reported using a wired or wireless mouse to complete the study, followed by a trackpad (32.98%); 1% reported using a touchscreen. In Study 1B, most participants (70.65%) reported using a mouse, followed by a trackpad (23.91%), a touchscreen (4.45%), or other device (1%), though participants selecting “other” did not provide additional details in the follow-up response.

No participants in either Study 1A or Study 1B reported encountering technical issues. In Study 1A, no participants reported finding the study confusing or tedious. In Study 1B, no participants reported the study as confusing, and two participants (2%) reported that the study was tedious. In their open-ended responses, these participants noted that the task required sustained focus and felt repetitive. No participants in Study 1A reported using external sources to complete the study; in Study 1B, 1% reported using an external source but did not provide further details in the follow-up.

Figure SM4 presents plots of ranking behavior in Study 1B, and details of the regression analyses testing the 3F pattern are provided in the section below.

Figure SM4*Steps of Ranking Shapes by Color (A) and Corners (B) in Study 1B*

Note. Mean rank of each shape across drag-and-drop steps in color darkness (Panel A) and number of corners (Panel B) tasks. Error bars represent ± 1 standard error across respondents at each step. The first four drag-and-drop actions are displayed for each respondent. Mean rank changes reflect both active movements and passive rank shifts of items.

Regression Analyses for testing the 3F Pattern

To reflect the nature of the three process measures - drag count, drag order, and drag distance - we used different regression frameworks tailored to each outcome. Drag count was analyzed using a generalized linear model (`glm()` in R), drag order was analyzed using a cumulative link model (`clm()` in R), and drag distance was analyzed as a continuous outcome using linear regression (`lm()` in R). For each of the three drag and drop measures, we estimated three models: (1) a fixed-effects model, (2) a mixed-effects model with participant-level random intercepts, and (3) a mixed-effects model with both participant- and item-level random

intercepts. Although model (3) was preregistered, the data structure led to estimation issues for this specification (e.g., non-estimable adjusted R^2), indicating overparameterization. We therefore additionally evaluated the simpler models and compared all three specifications using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Model comparisons consistently favored simpler specifications, and results were substantively consistent across models. Specifically, the best-fitting models included no random intercepts for drag count or drag order, and only participant-level random intercepts for drag distance, as reported in Tables SM1 and SM2. Accordingly, we report the best-fitting models in the main text (e.g., using the respective coefficients in Figure 3). In descriptive figures, rank positions retain their conventional coding, with 1 indicating the highest or top position and 6 the lowest or bottom position. For the regression analyses below, we reverse-coded the attribute-rank predictors so that larger values indicate higher attribute values (6 = highest; 1 = lowest). We used the same regression-coding convention for the Study 2 attribute-ranking analyses.

Table SM1*3F Regression Models in Study 1A*

Predictors	Drag Count			Drag Order			Drag Distance		
	Log-Odds	std. Error	p	Log-Odds	std. Error	p	Estimates	std. Error	p
Corner Rank [Nested in Corner]	0.76	0.08	<0.001	-0.97	0.06	<0.001	0.54	0.02	<0.001
Corner Rank [Nested in Color]	0.24	0.07	<0.001	0.01	0.05	0.752	-0.02	0.02	0.195
Color Rank [Nested in Color]	0.87	0.09	<0.001	-1.08	0.06	<0.001	0.50	0.02	<0.001
Color Rank [Nested in Corner]	0.12	0.07	0.083	0.20	0.05	<0.001	0.04	0.02	0.029
Random Effects									
σ^2								0.55	
τ_{00}							0.20	ResponseId	
ICC							0.27		
N							91	ResponseId	
Observations		1056			1056			1056	
R ² Tjur		0.469			0.563			0.673 / 0.760	

Note. For ease of presentation, the table omits intercepts for the drag count and drag distance models, threshold parameters for the drag order model, controls for initial item positions, and the condition dummy. All models include these terms in the estimation. Predictors are reverse-coded, where 6 denotes the highest attribute value and 1 denotes the lowest attribute value. The regression models include 91 unique participants in total. Of these, 87 participants provided valid responses in the color task and 89 provided valid responses in the corner task, resulting in 1,056 total observations (six observations per person per task).

Table SM2*3F Regression Models in Study 1B*

Predictors	Drag Count			Drag Order			Drag Distance		
	Log-Odds	std. Error	p	Log-Odds	std. Error	p	Estimates	std. Error	p
Corner Rank [Nested in Corner]	0.43	0.07	<0.001	-0.75	0.06	<0.001	0.56	0.02	<0.001
Corner Rank [Nested in Color]	0.05	0.07	0.515	-0.11	0.05	0.028	-0.03	0.02	0.251
Color Rank [Nested in Color]	0.49	0.07	<0.001	-0.78	0.06	<0.001	0.58	0.02	<0.001
Color Rank [Nested in Corner]	0.04	0.07	0.546	-0.32	0.05	<0.001	-0.02	0.02	0.458
Random Effects									
σ^2								0.83	
τ_{00}							0.44	ResponseId	
ICC							0.35		
N							88	ResponseId	
Observations	960			960			960		
R ² Tjur	0.247			0.436			0.570 / 0.719		

Note. For ease of presentation, the table omits intercepts for the drag count and drag distance models, threshold parameters for the drag order model, controls for initial item positions, and the condition dummy. All models include these terms in the estimation. Predictors are reverse-coded, where 6 denotes the highest attribute value and 1 denotes the lowest attribute value. The regression models include 88 unique participants in total. Although each task has 80 valid responses, the samples are not fully overlapping because DROPT measures could only be calculated reliably for one task for some participants. This results in 480 valid responses for each task and hence 960 valid responses in total.

Simulation Results of Alternative Sorting Strategies

Table SM4 describes each simulated sorting strategy and illustrates how each strategy transforms the same initial order into the same target order.

Table SM3*Simulated Sorting Strategies and Example Sorting Paths*

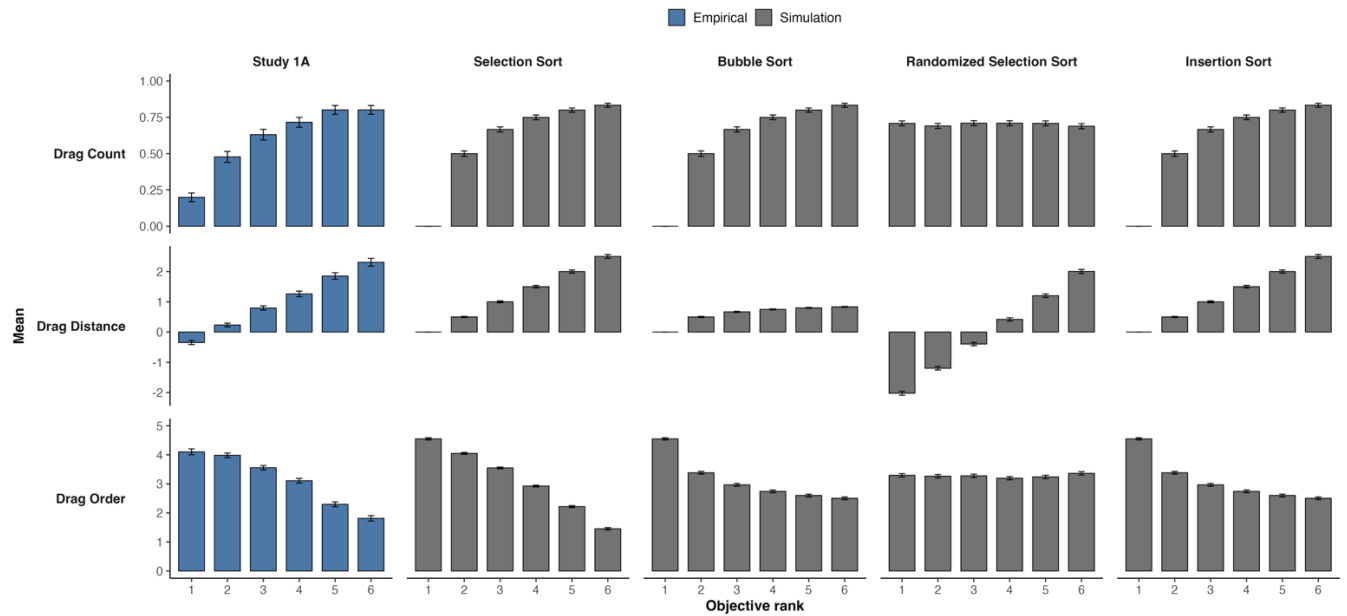
Algorithm	Sorting Procedure	Example sorting path: 163542 → 123456
Selection sort	At each step, the sorter identifies the highest-value remaining item and moves it into its correct position.	Move item: 2 → New order: 126354 Move item: 3 → New order: 123654 Move item: 4 → New order: 123465 Move item: 5 → New order: 123456
Bubble sort	The sorter scans the list from top to bottom and corrects adjacent inversions one at a time, repeatedly, until the list is sorted.	Move item: 3 → New order: 136542 Move item: 5 → New order: 135642 Move item: 4 → New order: 135462 Move item: 2 → New order: 135426 Move item: 4 → New order: 134526 Move item: 2 → New order: 134256 Move item: 2 → New order: 132456 Move item: 2 → New order: 123456
Randomized selection sort	At each step, the sorter randomly selects one item that is out of order and drags it into its correct position.	Move item: 5 → New order: 163452 Move item: 6 → New order: 134526 Move item: 2 → New order: 123456
Insertion sort	The sorter scans the list from top to bottom, treating the first item as already organized, and moves each subsequent item into its appropriate position relative to the items already considered.	Move item: 3 → New order: 136542 Move item: 5 → New order: 135642 Move item: 4 → New order: 134562 Move item: 2 → New order: 123456

Note. The table illustrates how each sorting strategy transforms the same initial order, 163542, into the same target order, 123456. 1 denoting the highest-value item and 6 denoting the lowest-value item. Each line in the example sorting path corresponds to one drag-and-drop movement. The moved item and resulting order are shown for each

step. Because randomized selection sort involves random choice, the path shown is one possible example rather than a unique sequence.

Beyond inspecting mean rank trajectories over steps, we also examined other drag-and-drop measures generated by each alternative algorithm. These comparisons suggest that the proposed selection-sort process provides the closest fit to the empirical pattern.

Across studies, participants typically completed the ranking task in about four steps. Bubble sort clearly deviated from this benchmark, requiring an average of 7.51 steps, substantially more than observed empirically. It also produced much smaller drag distances than the empirical data, because all movements in bubble sort are adjacent one-step swaps. As a result, the average drag distance for each item was below 1 (see Figure SM5).

Figure SM5*Drag-and-Drop Measure Means by Item in Study 1A and Simulations*

Note. Study 1A data are collapsed across color and corner tasks. Bars show means, and error bars show standard errors, for items ordered by objective rank from 1 (highest value) to 6 (lowest value). Positive drag-distance values indicate upward movements in the ranking list, whereas negative values indicate downward movements.

For randomized selection sort, because the item selected at each step is random, the algorithm does not produce a systematic relationship between item rank and either drag count or drag order. As shown in Figure SM5, the drag-count and drag-order bars are essentially flat across item ranks. Randomized selection sort also produces many downward movements, indicated by negative drag-distance values, which is inconsistent with the empirical pattern.

For insertion sort, the drag-order pattern is flatter and shows less of a prioritization trend than the empirical data. This difference is especially clear in a diagnostic case: when the highest-ranked item begins in the last position. The proposed selection sort predicts that this item should still be moved early, often first, because the process prioritizes the highest-ranked remaining

item and moves it to the top. By contrast, insertion sort predicts that this item should be moved later, because the process scans sequentially through the list. Consistent with the selection-sort account, in this subset of the empirical data, the highest-ranked item was moved relatively early, with an average drag order of 1.69. This was closer to the selection-sort benchmark of 1.00 than to the insertion-sort benchmark of 3.72. This analysis included 32 observations across tasks in Study 1A and 120 observations across the simulations.

Finally, we examined correlations among the three drag-and-drop measures generated by each simulated algorithm, using the same calculation approach for study 1A (see Table SM4).

Table SM4

Mean Correlation Among Drag-and-Drop Measures in Study 1A and Sorting Simulations

	Order and distance	Count and distance	Count and order
Study 1A	-0.67 (0.37)	0.49 (0.37)	-0.73 (0.17)
Selection Sort	-0.88 (0.08)	0.61 (0.28)	-0.70 (0.23)
Bubble Sort	-0.92 (0.10)	1.00 (0.00)	-0.92 (0.10)
Randomized Selection Sort	-0.00 (0.42)	-0.01 (0.38)	-0.68 (0.30)
Insertion Sort	-0.49 (0.27)	0.61 (0.28)	-0.92 (0.10)

Note. Study 1A data were collapsed across tasks for this calculation. For Study 1A, values indicate mean within-participant correlations between pairs of drag-and-drop measures. For simulations, values indicate mean correlations computed across initial orders. Standard deviations are shown in parentheses

In addition, the simulations show that the three measures are not necessarily correlated simply because they come from the same task. Under randomized selection sort, for example, the correlations involving drag distance were near zero: order and distance were uncorrelated, and count and distance were also close to zero. This shows that the extent to which the three drag measures overlap is a function of the strategy used by sorters.

Study 2 Supplementary Methods

See Table SM5 for lottery stimuli details in the attribute ranking tasks, which are identical to those in the preference ranking task for each participant. The interface participants faced in the attribute ranking tasks (i.e., rank by probability and rank by payoff) were very similar to those in Study 1, except that participants now ranked lotteries varying in amount and probability. For conciseness, the interface is not included here; the full Qualtrics materials are available on OSF. Similar to Studies 1A and 1B, in Study 2, participants were asked to recall the ranking instructions after completing the attribute-ranking and preference ranking tasks by selecting three statements that best described how they completed the task. We report the results in the supplementary section below.

Preference and Validation Ranking Task

Table SM5 presents the stimuli in the preference and validation ranking tasks. Figure SM6 presents the task interface for the preference ranking task, in which participants were asked to rank lotteries based on how much they liked each one. The interface used in the validation ranking task was very similar and is not shown here for conciseness; full Qualtrics materials are available on OSF.

The validation ranking task was incentivized with bonus monetary reward. Before beginning the task, participants read the following instructions:

You have the chance to earn a bonus in the upcoming task. If you win, the bonus will be distributed to your Prolific account within 24 hours after you complete the study. To determine your bonus, two of the six lotteries from the upcoming task will be randomly selected, and the lottery you rank higher (i.e., the one you would like to play more) will be played. Because any pair of lotteries could be selected (e.g., 1st vs. 2nd or 5th vs. 6th)

to determine your bonus, it is important to rank all six lotteries in the order you would like to play them.

Table SM5

Stimuli in the Preference and Validation Ranking Tasks

Task	Lottery Number	Probability (%)	Payoff (\$)	Mean Rank
Preference Ranking Task	1	94 (5)	2.90 (1.25)	1.42 (1.95)
	2	54 (18)	5.40 (1.25)	1.82 (1.45)
	3	29 (6)	9.70 (3.05)	2.52 (1.03)
	4	17 (6)	17.50 (4.75)	2.77 (1.00)
	5	9 (2)	31.50 (9.25)	3.20 (1.44)
	6	5 (2)	56.70 (15.95)	3.28 (2.17)
Validation Ranking Task	1	84 (10)	3.40 (0.4)	1.60 (2.02)
	2	63 (11)	4.40 (0.4)	1.79 (1.44)
	3	31 (8)	9.10 (4.3)	2.27 (1.12)
	4	15 (8)	18.70 (5.3)	2.64 (1.10)
	5	6 (1)	47.70 (23.7)	3.27 (1.44)
	6	3 (2)	94.40 (22)	3.41 (2.01)

Note. Values in parentheses indicate the maximum absolute jitter added for lottery. For example, the probability for Lottery 1 was drawn from a uniform distribution on $[84 - 10, 84 + 10]$ and then rounded to the nearest integer; the payoff for Lottery 1 was drawn from $[3.40 - 0.40, 3.40 + 0.40]$ and rounded to two decimal places. Mean rank reports the average ranking across participants, with standard errors in parentheses. Smaller values indicate higher preference (1 = most preferred; 6 = least preferred).

Figure SM6*Task Interface for the Preference Ranking Task*

Please rank the following lotteries based on **how much you would like to play each one**.

To rank them, click and drag each lottery into order from the one **you would most like to play (at the top)** to the one **you would least like to play (at the bottom)**.

Remember, lotteries you would like to play more should be placed **above** lotteries you would like to play less. Please ensure that **every** lottery is in the right place.

Example: For instance, if you would like to play

A
xx%
\$yy

 more than

B
xx%
\$yy

, lottery A should be placed **above**.

34% \$11.18
18% \$21.43
64% \$6.01
4% \$52.63
9% \$28.35
93% \$3.93

Note. The black outline is for visual clarity in the SM only and was not shown to participants. The numerical values shown represent one specific realization of the bounded random jitter applied to the lottery parameters.

Table SM6 presents the model specifications for the four models discussed in the main text. All models included participant and item random intercepts and were estimated using the `clmm()` function in R for ordinal outcomes with mixed effects. The *baseline model* predicts final ranks as a function of probability and payoff using a single fixed effect for each attribute. The *individual random effects* model extends this by allowing the probability and payoff coefficients

to vary across participants. The *distance-weight augmented* model incorporates process-based attribute weights. These weights were extracted from the preference task using a linear mixed model that regressed each participant's residualized drag distances onto the item's probability and payoff, yielding participant-specific weights on probability and payoff. To obtain the residualized metric, the raw drag distance value was regressed on the item's starting location while including participant and item random intercepts, thereby removing systematic variation attributable to initial positions, participant-level tendencies, and item-specific characteristics. These participant-level weights were then interacted with the corresponding attributes in the ranking model. Finally, the *distance augmented* model adds each item's residualized drag distance directly as a predictor. Model fit indices for all four models are presented in Table 2 of the main text.

Table SM6*Mixed-Effects Model Specifications for the Preference Ranking Task*

Model	Latent Variable Equation	Notation
Baseline	$y_{ij}^* = \beta_{prob}x_{prob,ij} + \beta_{amt}x_{amt,ij} + u_j + v_i + \varepsilon_{ij}$	y_{ij}^*: latent utility for item i for participant j. $x_{prob,ij}, x_{amt,ij}$: probability and payoff attribute values (z-scored). $\beta_{prob}, \beta_{amt}$: fixed effect coefficients. $u_j \sim N(0, \sigma_{participant}^2), v_i \sim N(0, \sigma_{bet}^2)$: participant and item random intercepts. $\varepsilon_{ij} \sim \text{Logistic}(0,1)$: error term following a logistic distribution.
Individual Random Effects	$y_{ij}^* = (\beta_{prob} + b_{prob,j})x_{prob,ij} + (\beta_{amt} + b_{amt,j})x_{amt,ij} + u_j + v_i + \varepsilon_{ij}$	$b_{prob,j} \sim N(0, \sigma_{prob}^2), b_{amt,j} \sim N(0, \sigma_{amt}^2)$: participant-specific random slopes for probability and payoff.
Distance-Weight Augmented Random Effects	$y_{ij}^* = (\beta_{prob} + b_{prob,j})x_{prob,ij} + (\beta_{amt} + b_{amt,j})x_{amt,ij} + \beta_{dist(prob)}x_{dist(prob),j}x_{prob,ij} + \beta_{dist(amt)}x_{dist(amt),j}x_{amt,ij} + u_j + v_i + \varepsilon_{ij}$	$x_{dist(amt),j}, x_{dist(prob),j}$: participant j's weighting coefficients for probability and payoff estimated from residualized drag distance. $\beta_{dist(prob)}, \beta_{dist(amt)}$: fixed-effect coefficients.
Distance Augmented Random Effects	$y_{ij}^* = (\beta_{prob} + b_{prob,j})x_{prob,ij} + (\beta_{amt} + b_{amt,j})x_{amt,ij} + \beta_{dist}x_{dist,ij} + u_j + v_i + \varepsilon_{ij}$	β_{dist}: fixed-effect coefficient. $x_{dist,ij}$: the residualized drag distance of item i for participant j.

Note. All models were estimated using cumulative link mixed models (clmm()) function from the ordinal package in

R). Model equations are specified on a latent utility scale y_{ij}^* . In a cumulative link model, the observed ordinal outcome is assumed to arise from thresholding this latent utility: the observed final rank $y_{ij} \in \{1, \dots, 6\}$ equals category k when $\theta_{k-1} < y_{ij}^* \leq \theta_k$, where $\{\theta_k\}$ are estimated cutpoints.

Sensitivity Analysis

To assess whether the individual random-effect model could meaningfully recover heterogeneity in payoff and probability weighting despite the negative correlation between these attributes, we conducted a simulation sensitivity analysis. We simulated data for 100 participants ranking six lotteries. The simulated lotteries used the same base values and jitter values as in Study 2 and preserved the passive repositioning feature of the drag-and-drop interface. Each participant was assigned a payoff preference weight drawn from a uniform distribution between 0 and 1, and their probability preference weight was set to one minus their payoff weight. Thus, some simulated participants placed greater weight on payoff, whereas others placed greater weight on probability.

For each participant-by-lottery observation, we generated a latent utility score as a weighted combination of standardized payoff and standardized probability, plus an item-level random effect and random noise:

$$U_{ij} = \beta_{prob,i} x_{prob,ij} + \beta_{amt,i} x_{amt,ij} + \alpha_j + \varepsilon_{ij}$$

Here, $\beta_{prob,i}$ and $\beta_{amt,i}$ denote participant i 's simulated probability and payoff weights, respectively; $x_{prob,ij}$ and $x_{amt,ij}$ denote standardized lottery attributes; α_j denotes an item-level random effect; and ε_{ij} denotes random noise. Item random effects were drawn from a normal distribution with mean 0 and standard deviation 0.32. This allowed lotteries to vary modestly in baseline attractiveness beyond their payoff and probability values. For example, if lottery A had an initial utility score of 0.50 and lottery B had a score of 0.20, item-level variation of this magnitude could sometimes reverse this relative preference.

Noise was drawn from a normal distribution with mean 0. The noise variance was set relative to the variance of the systematic component of the latent score, producing approximate

R^2 values of .50, .40, and .30 across simulation conditions. These conditions allowed us to examine recovery under varying levels of noise.

The resulting latent utility scores were converted into within-participant ranks through an ordered ranking procedure, with ties broken randomly. We then fitted the same cumulative link mixed model used in Study 2 for individual random-effects model.

The key recovery statistic was the correlation between each participant's true assigned preference weights and their recovered participant-specific slopes from the model. We examined recovery for payoff weights, probability weights, and the relative weighting of payoff versus probability.

The results suggested that even under the noisiest simulation condition, corresponding to an approximate R^2 of .30, the model recovered meaningful attribute-specific heterogeneity. Across 100 simulations, participants with higher true payoff weights tended to have higher recovered payoff slopes, mean $r = .67$, and participants with higher true probability weights tended to have higher recovered probability slopes, mean $r = .69$. The relative weighting of payoff versus probability showed the same pattern: the mean correlation between true and recovered relative weights was $r = .72$. This suggests that the high correlation between probability and payoff does not necessarily preclude meaningful recovery of relative preference weights.

Validation Choice Task

Table SM7 presents the five binary lottery choices participants completed. All lotteries had roughly the same expected value. The two probability and payoff attributes were negatively correlated at $r = -0.93$.

Table SM7*Validation Choice Task Stimuli*

Choice Number	Lottery 1	Lottery 2	P-bet Chosen (%)
1	23% \$12.17	49% \$5.72	71.98%
2	77% \$3.64	24% \$11.67	64.29%
3	83% \$3.38	95% \$2.95	65.38%
4	63% \$4.44	55% \$5.09	59.89%
5	19% \$14.74	61% \$4.59	68.13%

Note. Participants chose between Lottery 1 and Lottery 2 on each trial. Both the left–right positions on the screen in which the two lotteries were presented and the order of the 5 trials were randomized. The exact wording of the question was: “Consider the following two lotteries. Which one would you like to play more?” The P-Lottery chosen (%) column reports the percentage of participants with complete and valid data for both preference tasks ($N = 182$) who selected the lottery with the higher probability of winning within each pair.

Study 2 Supplementary Results

After initial exclusions, the median education level was a bachelor’s degree (4-year), and the median income category was \$50,000–\$74,999. In Study 2, we asked participants whether they believed they had a real chance to earn a bonus, and the vast majority responded affirmatively (95.79%). In addition, 96.84% of participants correctly recalled the three tasks they completed. We preregistered exploratory moderation analyses using these variables, but because fewer than 5% of participants failed the relevant checks, yielding insufficient variability, we did not pursue these analyses.

We also summarize responses to the additional debrief questions for Study 2. Most participants (68.95%) reported using a wired or wireless mouse to complete the study, followed

by a trackpad (31.05%); 2.11% reported using a touchscreen. Two participants (1.05%) reported encountering technical issues; in follow-up responses, they indicated that they had to restart the survey at some point. These data were retained.

10% of participants reported that the study was tedious, and 3.16% reported that it was confusing. In follow-up open-ended responses, participants who found the study tedious noted that the task required sustained focus and felt repetitive, whereas those who reported confusion indicated that they found the ranking tasks unintuitive. All data was retained. No participants reported using external sources to complete the study. In addition, as a robustness check, we examined respondents flagged by Qualtrics as likely fraudulent (Fraud Score > 30; 8.76%). Unless otherwise noted, all results reported in the manuscript remain consistent when these participants are excluded.

Table SM8 and Figure SM7 report regression results from the three drag-and-drop models. In Study 2, the expected values of the lotteries were held approximately constant while probability and payoff varied. Hence, probability and payoff were highly correlated ($r = -0.71$), which prevents the regression models from cleanly identifying attribute usage. We believe that the observed effects of the task-irrelevant attribute more likely reflect stimulus-induced collinearity and resulting modeling constraints, rather than meaningful reliance on that attribute during the ranking process.

One reason for this interpretation is the cognitive demands such a strategy would entail. For example, in the payoff-ranking task, relying on probability would require participants to first recognize the negative relationship between payoff and probability and then mentally invert that relationship when identifying the next-best option at each step—a cognitively effortful strategy given that the task-relevant payoff information is directly available.

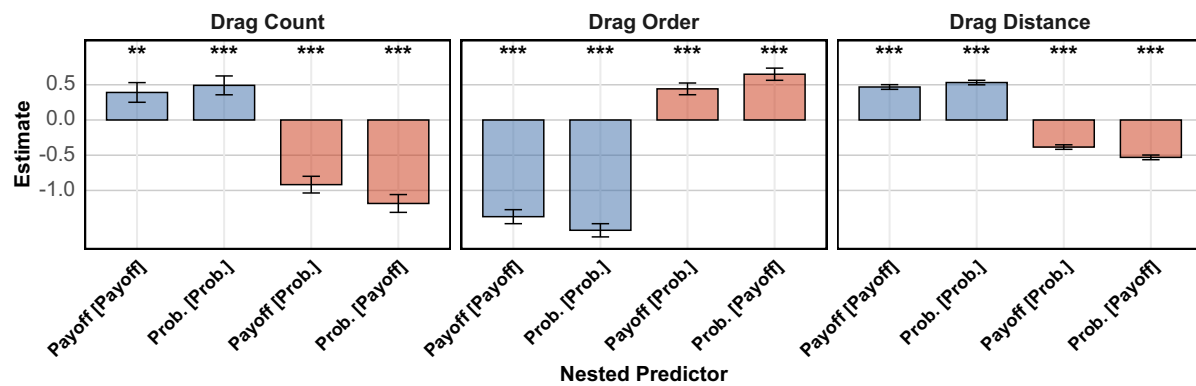
Table SM8[illegible]

	Drag Count	Drag Order	Drag Distance
τ_{00}			0.23 ResponseId
ICC			0.29
N			190 ResponseId
Observations	2130	2130	2130
R ² Tjur	0.468	0.582	0.671 / 0.768

Note. For ease of presentation, the table omits intercepts for the drag count and drag distance models, threshold parameters for the drag order model, controls for initial item positions, and the condition dummy. All models include these terms in the estimation. Predictors are reverse-coded, where 6 denotes the highest attribute value and 1 denotes the lowest attribute value.

Figure SM7

Regression Coefficients for Focal Predictors of Drag-and-Drop Behavior for Study 2



Note. Model-estimated coefficients (± 1 SE) across the three drag-and-drop regression models. All predictors are standardized (z-scored) prior to analysis. Before standardization, the attribute-rank predictors were reverse-coded so that larger values indicate higher attribute values (6 = highest; 1 = lowest). Under this coding, the 3F pattern appears as a positive coefficient for drag count, a negative coefficient for drag order, and a positive coefficient for drag distance. Each predictor refers to an attribute nested within a task, with the task indicated in brackets (e.g., “Payoff [Payoff]” indicates payoff attribute value nested in the rank-by-payoff task). Relevant predictors are shown in blue and irrelevant predictors in red. Asterisks above each bar denote significance: *** $p < .001$; ns = not significant.

Validation Ranking Task Supplementary Results

As discussed in the main text, we used the models trained on the preference ranking task to generate predictions for a subsequent, independent validation ranking task. For each model, we substituted the probability and payoff values, and the residualized drag distance when applicable, from the validation task into the model equations to obtain predicted latent preference scores. Lottery random intercepts were omitted during prediction as these terms were estimated using the specific items from the preference task and do not generalize to the new items shown in the validation task.

To statistically evaluate whether incorporating drag distance information improved prediction accuracy, we compared each drag distance model against the *individual random effects* model. We conducted both paired t-tests and paired Wilcoxon signed-rank tests on r Kendall's τ and Pearson's r . Pearson's r correlations were Fisher z -transformed prior to analysis. The *distance augmented* model significantly outperformed the *individual random effects* model across both fisher-transformed Pearson's r , $t(181) = 8.32, p < .001, V = 3018, p_{\text{Wilcoxon}} < .001, p_{\text{permutation}} < .001$, and Kendall's τ , $t(181) = 7.12, p < .001, V = 875, p_{\text{Wilcoxon}} < .001$. The *distance weight augmented* model improved significantly in Pearson's r , $t(181) = 4.46, p < .001, V = 5751, p_{\text{Wilcoxon}} < .001$, but showed a mixed pattern of statistical significance in Kendall's τ , $t(181) = 1.66, p = .098, V = 143, p_{\text{Wilcoxon}} = .038$. When individuals flagged as fraudulent by Qualtrics were excluded, neither test was significant for Kendall's τ ($ps > .10$).

Within each cross-validation split, we also computed held-out predictive accuracy separately for participants with mixed preference profiles and for participants whose choices were fully aligned with a pure P-bet or \$-bet preference. To classify participants, we computed

the Kendall's τ correlation between each participant's final ranking and a perfectly ordered P-bet or \$-bet ranking (e.g., ranking the highest-probability option first, second-highest second, and so on). Individuals with a correlation of 1 were classified as P- or \$-bet group participants ($N=91$), and all others were classified as having mixed preferences ($N=91$). This strategy differs from the top-choice classification used in the main text in that it yields a more balanced distribution across categories and provide a more exhaustive characterization of preferences by considering the full rank rather than the single top-ranked option. As shown in Table SM9, performance gains were larger for participants with mixed preference profiles.

Table SM9*Subgroup Comparison of Prediction Accuracies in the Validation Ranking Task*

Subgroup	Accuracy Metrics	Baseline	Individual Random Effects	Distance-Weight-Augmented Random Effects	Distance-Augmented Random Effects
P- or \$-bet group	Kendall's τ	$M = 0.47$ [0.31, 0.64]	$M = 0.76$ [0.65, 0.87]	$M = 0.76$ [0.65, 0.87]	$M = 0.79$ [0.71, 0.87]
	Pearson's r	$M = 0.48$ [0.32, 0.64]	$M = 0.76$ [0.65, 0.87]	$M = 0.77$ [0.66, 0.87]	$M = 0.82$ [0.73, 0.90]
Mixed preference group	Kendall's τ	$M = 0.28$ [0.16, 0.41]	$M = 0.36$ [0.25, 0.47]	$M = 0.40$ [0.29, 0.50]	$M = 0.71$ [0.66, 0.77]
	Pearson's r	$M = 0.32$ [0.19, 0.46]	$M = 0.43$ [0.31, 0.55]	$M = 0.45$ [0.33, 0.57]	$M = 0.78$ [0.73, 0.84]

Note. Pearson's r reflects the correlation between each participant's predicted latent utility scores and their observed final rankings. Kendall's τ reflects the correlation between predicted and observed rank orders after converting predicted latent utilities into predicted rankings. The table reports the mean and the 95% confidence interval. ($N = 91$ each for each subgroup).

Validation Choice Task Supplementary Results

To examine whether process measures extracted from the preference ranking task can improve predictions of binary choices, we compared several logistic regression models predicting participants' choices (see Table SM10).

Model 1 was a baseline model that included only the probability and payoff differences between the two lotteries. Model 2 augmented this baseline by weighting each attribute difference using participant-specific attribute weights estimated from final rankings. Model 3 further added participant-specific attribute weights estimated from residualized drag distance in the preference ranking task. This model tests whether DROPT process measures improve prediction beyond final rankings, without requiring process data from the validation choice task itself. Finally, Model 4 provided a non-DROPT benchmark by adding participants' self-reported risk preference to the final-rank-weight model in an analogous manner. This benchmark allowed us to assess whether the predictive value of DROPT measures exceeded that of a conventional individual-difference measure.

Table SM10*Model Specifications for the Validation Choice Task*

Model	Latent Variable Equation	Notation
Baseline (M1)	$y_{ij}^* = \beta_0 + \beta_{amt}x_{diff(amt),ij} + \beta_{prob}x_{diff(prob),ij} + u_j + v_i + \varepsilon_{ij}$	y_{ij}^*: latent utility for choosing the P-bet in lottery pair i for participant j. $x_{diff(amt),ij}$, $x_{diff(prob),ij}$: differences in probability and payoff between the two lotteries in pair i (z-scored). β_{amt}, β_{prob}: fixed-effect coefficients. u_j: participant random intercept. v_i: lottery-pair random intercept.
Final-Rank-Weight Augmented (M2)	$y_{ij}^* = \beta_0 + \beta_{rank(amt)}x_{rank(amt),j}x_{diff(amt),ij} + \beta_{rank(prob)}x_{rank(prob),j}x_{diff(prob),ij} + u_j + v_i + \varepsilon_{ij}$	$x_{rank(amt),j}$, $x_{rank(prob),j}$: participant j's weighting coefficients for probability and payoff estimated from final ranks. $\beta_{rank(amt)}$, $\beta_{rank(prob)}$: fixed-effect coefficients.
Final-Rank and Distance Weights Augmented (M3)	$y_{ij}^* = \beta_0 + \beta_{rank(amt)}x_{rank(amt),j}x_{diff(amt),ij} + \beta_{rank(prob)}x_{rank(prob),j}x_{diff(prob),ij} + \beta_{dist(amt)}x_{dist(amt),j}x_{diff(amt),ij} + \beta_{dist(prob)}x_{dist(prob),j}x_{diff(prob),ij} + u_j + v_i + \varepsilon_{ij}$	$x_{dist(amt),j}$, $x_{dist(prob),j}$: participant j's weighting coefficients for probability and payoff estimated from residualized drag distance. $\beta_{dist(amt)}$, $\beta_{dist(prob)}$: fixed-effect coefficients.
Final-Rank-Weight and Risk-Preference Augmented (M4)	$y_{ij}^* = \beta_0 + \beta_{rank(amt)}x_{rank(amt),j}x_{diff(amt),ij} + \beta_{rank(prob)}x_{rank(prob),j}x_{diff(prob),ij} + \beta_{risk(amt)}x_{risk,j}x_{diff(amt),ij} + \beta_{risk(prob)}x_{risk,j}x_{diff(prob),ij} + u_j + v_i + \varepsilon_{ij}$	

Note. All models were estimated as mixed-effects logistic regressions with a logit link. Model equations are written on a latent continuous scale y_{ij}^* , where the linear predictor corresponds to the log-odds of choosing the P-bet. The observed binary choice y_{ij} (1 = choosing the P-Lottery, 0 otherwise) is modeled as $\Pr(y_{ij} = 1) = \{1 + \exp(-y_{ij}^*)\}^{-1}$.

We evaluated out-of-sample predictive performance by training each model on 80% of participants and assessing performance on the remaining 20%. This procedure was repeated 5,000 times, with participants being randomly assigned to the training or held-out sample in each iteration, to obtain precise performance estimates. Table SM11 presents the results. Within each iteration, we additionally computed held-out predictive accuracy separately for individuals classified as P- or \$- bet group participants and mixed preference participants. We found a modest improvement in prediction accuracy when adding process-based weights (i.e., comparing M3 vs. M2), with larger gains observed among participants with mixed preferences.

Table SM11*Held-Out Prediction Accuracy Across Models*

Model	M1	M2	M3	M4
Full held-out sample	66.03% (5.14%)	74.41% (4.35%)	74.98% (4.30%)	71.39% (4.62%)
P- or \$- bet group	69.51% (8.00%)	81.32% (6.21%)	81.30% (6.21%)	77.49% (6.53%)
Mixed preference group	62.54% (6.71%)	67.53% (5.80%)	68.70% (5.80%)	65.30% (6.46%)

Note. The table reports mean held-out accuracy across 5,000 iterations, with standard deviations shown in parentheses. Boldface denotes the model with the highest mean held-out accuracy within each subgroup.

Summary of Pre-Registered Hypotheses Tests

Across all three studies, we predicted that participants would (H1) drag and drop items ranking higher on the relevant attribute more frequently, (H2) move them earlier in the ranking process, and (H3) move them greater distances upward. Evidence for these hypotheses is

provided by the regression coefficients summarized in Figures 3 (Studies 1A and 1B) and SM7 (Study 2) and reported in detail in the regression models presented in Tables 1,2 and 8 of the SMs. Across studies 1A-1B, all task-relevant nested predictors are statistically significant ($p < .05$) and larger in magnitude than their corresponding non-task-relevant predictors. In Study 2 attribute ranking tasks, coefficients were more similar in magnitude across some measures, which likely reflect the high correlation between attributes. For Studies 1A and 1B, we additionally pre-registered that (H4) most drag-and-drop actions (i.e., more than 50%) would result in an item being moved upward in the ranking. To test this hypothesis, we conducted a one-sample test of a proportion against a null of 0.5 in studies 1A and 1B in the color and corner ranking tasks, respectively. Table SM12 provides evidence for this hypothesis.

Table SM12

Proportion of Upward Drag-and-Drop Actions in Studies 1A and 1B (H4)

Study (Task)	Proportion upward [95% Confidence Interval]	N	p
1A (Color)	0.93 [0.90, 1.00]	315	< .001
1A (Corner)	0.86 [0.83, 1.00]	323	< .001
1B (Color)	0.77 [0.72, 1.00]	295	< .001
1B (Corner)	0.80 [0.76, 1.00]	291	< .001

Note. The reported proportion reflects the share of drag-and-drop actions that moved an item upward in the ranking.

N corresponds to the total number of drag and drop actions.

* p-value results are based on a one-sided one-sample binomial test comparing the observed proportion against a null value of 0.5.

In Study 2, we predicted that considering participants' drag-and-drop pattern would improve predictions of their preferences in the validation ranking and choice tasks. We found evidence for both as detailed in the main text and *Study 2 Supplementary Results*.